23A01601 · DESIGN OF STEEL STRUCTURES

UNIT – I

Introduction to Structural Steel and Design of Connections

Syllabus: General – Types of Steel – Properties of Structural Steel – I.S. Rolled Sections – Concept of Limit State Design – Design of Simple and Eccentric Bolted and Welded Connections – Types of Failure and Efficiency of Joint – Prying Action – Introduction to HSFG bolts.

What's inside these notes

- Types of steel & properties (IS 2062 grades)
- IS rolled sections & section classification
- Limit State Design – safety factors, load combinations
- Bolted connections: bearing bolts, capacities, failures, efficiency
- Eccentric bolted connections (torsion + shear, bending + shear)
- Welded connections: fillet & butt, simple and eccentric
- Prying action · HSFG bolts · solved examples

Exam tip: Every connection problem = (1) find capacity of ONE bolt/weld, (2) divide load, (3) check plate. Write γ_m values clearly – marks are given for them.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2\times10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Introduction to Structural Steel

Steel = alloy of **iron + carbon (0.12 – 0.25 % for structural steel)** + small amounts of Mn, Si, S, P. Used for buildings, bridges, towers, industrial sheds, transmission lines, silos, chimneys etc.

Advantages

- **High strength-to-weight ratio** → light structures, smaller foundations
- **Ductile** – large deformation before failure (gives warning)
- Uniform quality, **fast erection**, prefabrication in shop
- Can be **reused / recycled**; easy to strengthen or modify later
- Long spans and slender members possible; good toughness & fatigue strength

Disadvantages

- Corrosion → painting / galvanising needed (maintenance cost)
- Loses strength at high temperature (≈ 50 % at 550 °C) → fire protection
- Slender members buckle → bracing needed; fatigue & brittle fracture at low temp.

Types of structural steel (IS 2062)

Type	Grade / designation	f_y (MPa)	f_u (MPa)
Mild steel	E250 (Fe 410)	250	410
High-tensile (HSLA)	E300 (Fe 440)	300	440
HSLA	E350 (Fe 490)	350	490
HSLA	E410 (Fe 540)	410	540
Very high strength	E450 (Fe 570)	450	570

> Quality grades A, B, C (BR, B0): differ in **killed/semi-killed**, impact test & weldability. Grade A – no impact test; B – 25 °C test; C – 0 °C. In E250 (Fe 410W) "W" = weldable.

Other types: **cast steel** (complex shapes, bearings), **stainless steel** (corrosion resistant), **weathering steel / COR-TEN** (forms protective rust layer), **cold-formed steel** (thin sections, IS 801), **wrought iron/cast iron** (obsolete).

2. Properties of Structural Steel

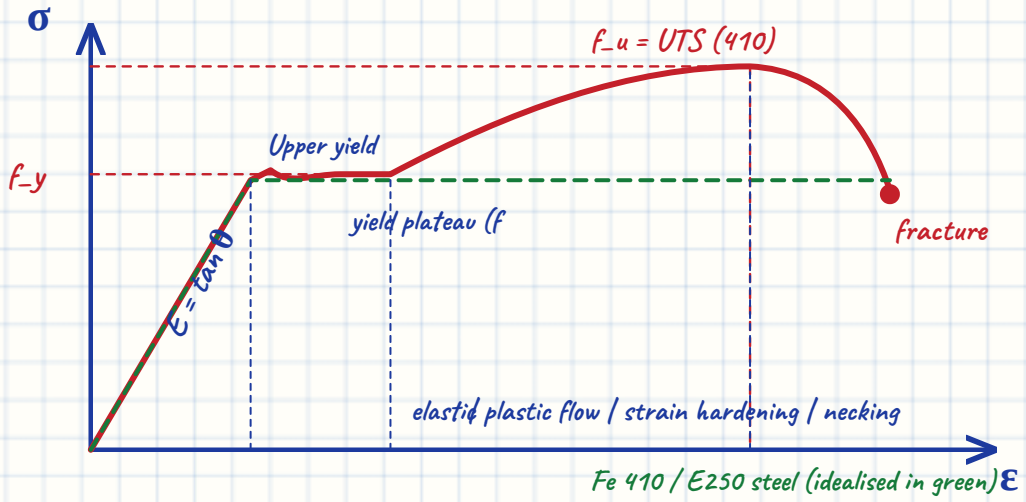
Physical (IS 800 – Cl. 2.2.4)

Property	Value
Unit weight (density)	7850 kg/m³ (78.5 kN/m³)
Modulus of elasticity E	2×10^5 MPa (2.0×10^5 N/mm²)
Shear modulus G	0.769×10^5 MPa
Poisson's ratio μ	0.3 (elastic range)
Coeff. of thermal expansion α	12×10^{-6} per °C

$G = E / [2(1+\mu)]$

Mechanical

- **Yield stress f_y** – design stress in tension/compression
- **Ultimate (tensile) strength f_u** – used for net-section & bolt checks
- **Ductility** – % elongation ≥ 23 % (gauge length $5.65\sqrt{A_0}$) for Fe 410
- **Toughness** – energy absorbed in Charpy test; resists brittle fracture
- **Hardness, weldability, fatigue strength**, residual stress from rolling/welding



Stress–strain curve of mild steel

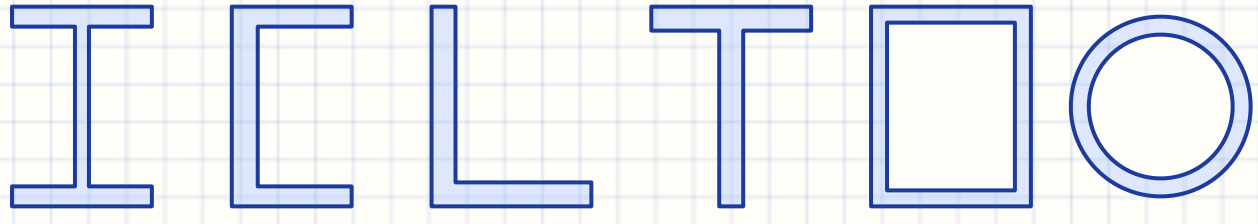
> Mild steel shows a **sharp yield point + yield plateau** ($\varepsilon \approx 0.2\ \% \rightarrow 2\ \%$), strain hardening up to f_u, then necking. High-tensile steel has **no clear yield point** → yield taken as **0.2 % proof stress**.

Why ductility matters

Ductile steel redistributes stress at holes/connections and allows **plastic analysis**; it warns before collapse.

3. I.S. Rolled Steel Sections

Hot-rolled sections are standardised (IS 808, properties in **SP-6 (1)** steel tables) and available in lengths up to 12–15 m.



Common rolled and hollow sections

Section	Designation example	Use
I-beams	ISMB 300 (medium weight), ISLB, ISJB (junior), ISWB (wide)	beams, columns
Heavy beam / column	ISHB 300	columns (bf ≈ depth)
Channels	ISMC 150, ISLC	purlins, built-up members
Angles	ISA 90×90×8 (equal), ISA 100×75×8 (unequal)	truss members, bracing
Tee	ISNT, ISST	chords of light trusses
Flats / plates	ISF 100×10	gussets, cover plates
Hollow	Tube (CHS), SHS, RHS (IS 4923)	space frames, columns

Designation: **ISMB 300 @ 44.2 kg/m** → I-section, Medium-weight Beam, depth 300 mm, mass 44.2 kg/m.

Section properties to read from tables

A (area), I_{zz}, I_{yy}, r_{zz}, r_{yy}, Z_{ez}, Z_{pz}, t_f, t_w, root radius R_1.

Example ISMB 300: A = 5626 mm², I_{zz} = 8603.6 cm⁴, I_{yy} = 453.9 cm⁴, Z_{ez} = 573.6 cm³, r_{yy} = 28.4 mm.

Section classification (IS 800 Table 2) – $\varepsilon = \sqrt{(250/f_y)}$

Class	Behaviour	Moment capacity
1 Plastic	forms plastic hinge + rotation capacity	$Z_p f_y/\gamma_{m0}$
2 Compact	reaches M_p, limited rotation	$Z_p f_y/\gamma_{m0}$
3 Semi-compact	reaches yield in extreme fibre only	$Z_e f_y/\gamma_{m0}$
4 Slender	local buckling before yield	effective section

Limits (rolled I): flange outstand $b/t_f \le 9.4\varepsilon$ (plastic)… 10.5ε (compact)… 15.7ε (semi-compact); web $d/t_w \le 84\varepsilon$ / 105ε / 126ε.

4. Concept of Limit State Design (LSM)

Limit state = condition beyond which the structure no longer satisfies the design requirement. IS 800:2007 is based on LSM (earlier IS 800:1984 used Working Stress).

Two groups

- **Limit State of Strength (collapse)** – yielding, rupture, buckling, fracture, fatigue, overturning, sway stability → use **factored loads**
- **Limit State of Serviceability** – deflection, vibration, durability, fire resistance, corrosion → use **unfactored (service) loads**

Design format

$$\text{Design action} \le \text{Design strength}$$

$$S_d \le R_d = R_u / \gamma_m$$

Partial safety factors apply on **loads (γ_f)** and on **material strength (γ_m)**.

Partial safety factors for materials (IS 800 Table 5)

Limit state	γ_m
Yielding, buckling (γ_{m0})	**1.10**
Ultimate stress / rupture (γ_{m1})	**1.25**
Bolts – bearing type (γ_{mb})	1.25
HSFG bolt – slip (γ_{mf})	1.25 (1.10 if slip is a serviceability limit)
Shop welds (γ_{mw})	**1.25**
Field welds (γ_{mw})	**1.50**

Load combinations for strength (IS 800 Table 4)

$$DL + LL \rightarrow 1.5\,DL + 1.5\,LL$$

$$DL + LL + WL/EL \rightarrow 1.2\,(DL + LL + WL)$$

$$DL + WL/EL \rightarrow 1.5\,DL + 1.5\,WL \text{ (and } 0.9\,DL + 1.5\,WL \text{ for uplift/stability)}$$

Accidental / crane loads – see code

Deflection limits (IS 800 Table 6, short list)

Member	Limit
Floor beams (supporting plaster, LL)	span/360
Other floor / roof members	span/300
Purlins, girts – elastic cladding	span/150
Purlins, girts – brittle cladding	span/180
Gantry girder – EOT crane (< 50 t)	span/750
Gantry girder – manual crane	span/500

Limit-state advantage: same reliability for all members, economical, treats loads and materials separately.

5. Bolted Connections – Basics

Connections join members and transfer force. **Bolts, rivets (obsolete), welds.**

Types of bolts

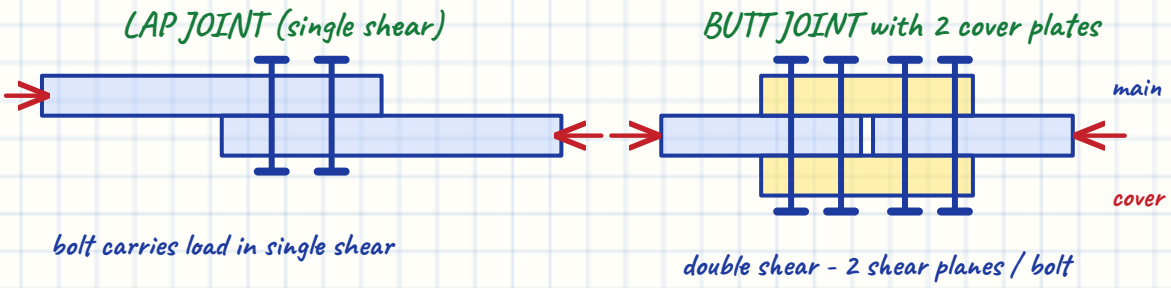
- **Black (unfinished) bolts** – grade 4.6, hole clearance, low cost – light structures, secondary members
- **High-strength bolts, bearing type** – grade 8.8 / 10.9 – shear carried by bolt in bearing
- **HSFG bolts** – tightened to give pre-tension; load transferred by **friction** (no slip)

Grade "4.6": $f_{ub} = 4 \times 100 = 400$ MPa, $f_{yb} = 0.6 \times 400 = 240$ MPa. Grade 8.8: $f_{ub} = 800$, $f_{yb} = 640$. Grade 10.9: $f_{ub} = 1000$, $f_{yb} = 900$.

Joint types

- **Lap joint** – plates overlap → single shear + eccentricity (bending)

- **Butt joint** – plates butt, cover plate(s) on one or both sides → single / **double shear**



Lap joint and Butt joint

Bolt hole diameter d_0 (IS 800 Table 19)

Bolt dia d	d_0
12 – 14 mm	d + 1 mm
16 – 22 mm	d + 2 mm
≥ 24 mm	d + 3 mm

Pitch and edge distance (IS 800 Cl. 10.2)

- **Min. pitch / gauge** $p, g \geq 2.5\ d$
- **Max. pitch** ≤ min(32 t, 300 mm) tension / compression-outer rows: 16 t or 200 mm tension; 12 t or 200 mm compression (t = thinner plate)
- **Min. edge/end distance** $e \geq 1.7\ d_0$ (rolled, machine flame-cut) ; $1.5\ d_0$ (sheared / hand flame-cut)
- **Max. edge distance** $\leq 12\ t\ \varepsilon$ ($\varepsilon = \sqrt{(250/f_y)}$)

> Why these limits? Small pitch → bearing/tear-out failure & hard to tighten; big pitch → plates gape, corrosion, buckling of compression plates.

6. Strength of a Bearing Bolt (IS 800 Cl. 10.3)

(a) Shear capacity

$$V_{dsb} = V_{nsb} / \gamma_{mb} ; V_{nsb} = (f_{ub}/\sqrt{3})\ (n_n A_{nb} + n_s A_{sb})$$

n_n = no. of shear planes **through threads**, n_s = through shank. A_{nb} = net tensile area at threads ≈ $0.78 \times \pi\ d^2/4$, A_{sb} = shank area $\pi\ d^2/4$.

Take $f_{ub}/\sqrt{3}$ and divide by $\gamma_{mb} = 1.25$.

(b) Bearing capacity (on plate)

$$V_{dpb} = V_{npb} / \gamma_{mb} ; V_{npb} = 2.5\ k_b\ d\ t\ f_u$$
$$k_b = \text{smaller of } e/(3d_0)\ ,\ p/(3d_0) - 0.25\ ,\ f_{ub}/f_u\ ,\ 1.0$$

(t = sum of thicknesses of plates bearing in the same direction; use f_u **of plate**.)

(c) Bolt value = min(shear, bearing)

Number of bolts $n = P_u$ / bolt value.

(d) Tension capacity of bolt

$$T_{db} = T_{nb}/\gamma_{mb} \; ; \; T_{nb} = 0.9\, f_{ub}\, A_{nb} \le f_{yb}\, A_{sb}\, (\gamma_{mb}/\gamma_{m0})$$

(e) Combined shear + tension

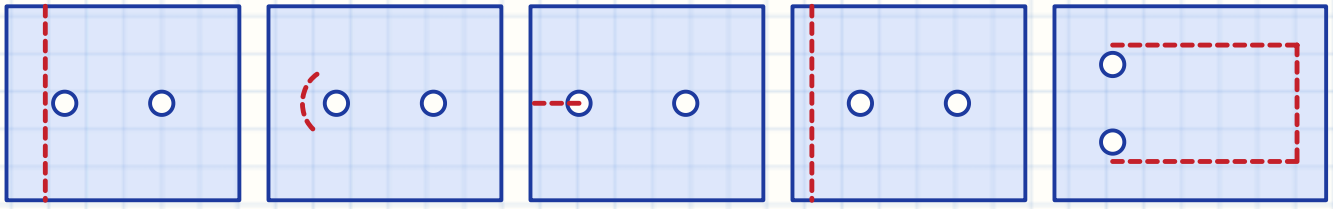
$$(V_{sb}/V_{db})^2 + (T_b/T_{db})^2 \le 1.0$$

Reduction factors

- **Long joint** ($l_j > 15\,d$): $\beta_{lj} = 1.075 - l_j/(200\,d)$ $(0.75 \le \beta_{lj} \le 1.0)$ – bolts at ends carry more
- **Large grip** ($l_g > 5\,d$): $\beta_{lg} = 8d/(3d + l_g)$
- **Packing plates** $t_{pk} > 6$ mm: $\beta_{pk} = 1 - 0.0125\, t_{pk}$

> Bolts in **double shear** have two shear planes → capacity ≈ 2× (but check bearing on the thinner plate set).

7. Types of Failure of a Bolted Joint



Failure modes of a bolted lap joint

Failure	Cause	Prevented by
1. Shear of bolt	bolt too small / few	more / bigger bolts, double shear
2. Bearing of plate (or bolt)	thin plate, small edge dist.	thicker plate, bigger d
3. Edge tear-out (shear-out)	edge/end distance too small	$e \ge 1.5$–$1.7\, d_0$
4. Tension failure of plate at net section	holes reduce area	adequate width, staggered holes
5. Block shear	tearing out a block of plate along bolt line	check T_{db} (see Unit II)
6. Bolt tension / prying	tension joints	thicker flange, more bolts
7. Buckling of plate	compression, large pitch	pitch limits

Net area of a plate

$$A_n = (b - n\,d_0)\,t \text{ (n holes in the section)}$$

For staggered holes: > $A_n = [\,b - n\,d_0 + \Sigma\, p^2/(4g)\,]\,t$

(p = stagger pitch, g = gauge between the two holes.)

Strength of plate

$$\text{Yielding of gross section: } T_{dg} = A_g\, f_y / \gamma_{m0}$$
$$\text{Rupture of net section: } T_{dn} = 0.9\, A_n\, f_u / \gamma_{m1}$$

Efficiency of a joint

$$\eta = \text{Strength of joint / Strength of solid plate} = (\text{min of } P_{ds}, P_{db}, P_{dn}) / (A_g\, f_y/\gamma_{m0})$$

Always < 100 %. Typical 60 – 85 %. Improve with double cover plate (double shear), staggered/zig-zag bolts.

8. Design of Simple Bolted Connection – Examples

EXAMPLE 1 – STRENGTH + EFFICIENCY OF A LAP JOINT

Two plates 100 × 10 mm (Fe 410) lapped, joined by **3 bolts M16, grade 4.6**, in one line along the load. e = 30 mm, p = 40 mm. Find joint strength & efficiency.

Bolt: d_0 = 18 mm, A_{sb} = 201 mm², A_{nb} = 0.78 × 201 = 157 mm².

Shear (single, thread in plane): V_{dsb} = (400/√3) × 157 / 1.25 = **29.0 kN**

Bearing: k_b = min(30/54 = 0.556, 40/54 − 0.25 = 0.49, 400/410 = 0.976, 1) = **0.49**

V_{dpb} = 2.5 × 0.49 × 16 × 10 × 410 /1.25 = 64.3 kN

Bolt value = 29.0 kN → 3 bolts : P_{ds} = **87.0 kN**

Plate: T_{dg} = 1000 × 250/1.1 = 227.3 kN ; A_n = (100−18)×10 = 820 mm² → T_{dn} = 0.9×820×410/1.25 = 242.1 kN

Strength of joint = **87.0 kN** (bolt shear governs)

$$\text{Efficiency } \eta = 87.0 / 227.3 = 0.38 = 38\ \%$$

EXAMPLE 2 – NUMBER OF BOLTS FOR A BUTT JOINT

Butt joint of 2 plates 200 × 12 mm with double cover plates 2 × 8 mm, M20 (4.6) bolts, to transmit **P_u = 300 kN**.

A_{sb} = 314, A_{nb} = 245 mm². **Double shear:** V_{dsb} = 2 × (400/√3) × 245 / 1.25 = **90.6 kN**

Bearing on main plate t = 12 (vs. 2 × 8 = 16 for covers → use 12): d_0 = 22 ; take e = 40, p = 60 : k_b = min(0.606, 0.659, 0.976, 1) = 0.606.

V_{dpb} = 2.5 × 0.606 × 20 × 12 × 410 / 1.25 = **119.3 kN**

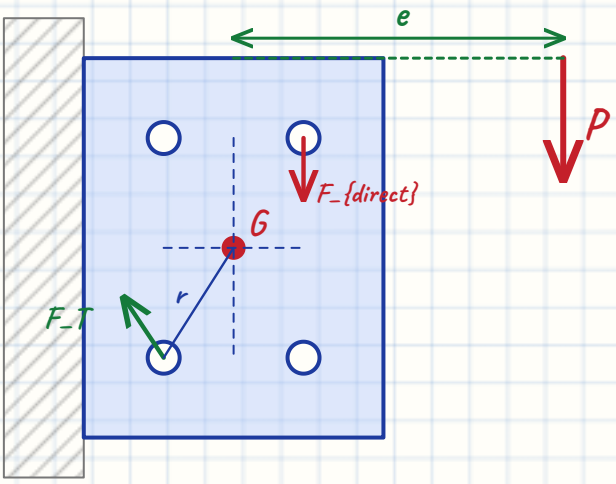
Bolt value = 90.6 kN → n = 300 / 90.6 = 3.3 → provide **4 bolts** on each side of the joint.

Check: $p \geq 2.5d = 50$ ✓, $e \geq 1.7\, d_0 = 37.4$ ✓.

9. Eccentric Bolted Connections

When the line of action of load does **not pass through the centroid of the bolt group** → extra moment $M = P \cdot e$.

Case A – Load in the plane of the bolt group (TORSION + direct shear)



Torsion $T = P \cdot e$

$F_T = T\, r / \Sigma r^2$

$F_d = P / n$

resultant = vector sum

Bracket connection – bolt group loaded eccentrically

Steps:

- Locate centroid G of bolt group.
- Direct shear on each bolt: $F_d = P/n$ (parallel to P).
- Torsion $T = P \cdot e$. Force on bolt i: $F_{T,i} = T\, r_i / \Sigma\, r_i^2$ ($\perp$ to r_i).
- Add vectors – resultant on the **worst bolt** $R = \sqrt{(F_x^2 + F_y^2)}$. Must be ≤ bolt value.

$$F_{T,i} = (P\, e)\, r_i / \Sigma (x^2 + y^2)$$

Components: horizontal = $F_T \cdot (y/r)$, vertical = $F_T \cdot (x/r)$.

EXAMPLE 3 – BRACKET WITH 4 BOLTS

Factored load P = 60 kN at e = 150 mm. 4 bolts at (±50, ±40) mm from G. Plate 10 mm, Fe 410, M22 (4.6) bolts.

$n = 4 \rightarrow F_d = 60/4 = 15$ kN (↓). $T = 60 \times 150 = 9000$ kN·mm

$r = \sqrt{(50^2 + 40^2)} = 64.03$ mm ; $\Sigma r^2 = 4 \times 4100 = 16400$ mm²

$F_T = 9 \times 10^6 \times 64.03 / 16400 =$ **35.1 kN**

Vertical comp. = 35.1 × 50/64.03 = 27.4 kN ; Horizontal = 35.1 × 40/64.03 = 21.95 kN

Critical bolt (both vertical parts add): V = 15 + 27.4 = 42.4 kN, H = 21.95 kN

$$R = \sqrt{(42.4^2 + 21.95^2)} = 47.8 \text{ kN}$$

Bolt capacity M22: $A_{nb} = 0.78 \times 380 = 296.5$; $V_{dsb} = (400/\sqrt{3}) \times 296.5 / 1.25 =$ **54.8 kN** > 47.8 ✓

Bearing: $e = 50$, $p = 80$, $d_0 = 24 \rightarrow k_b = \min(0.694, 0.861, 0.976, 1) = 0.694 \rightarrow V_{dpb} = 125.3$ kN ✓ → **Safe**

(M20 would give 45.3 kN < 47.8 → unsafe.)

Case B – Load perpendicular to plane of bolt group (TENSION + SHEAR)

Bracket bolted to column flange; vertical load P at eccentricity e from the flange face.

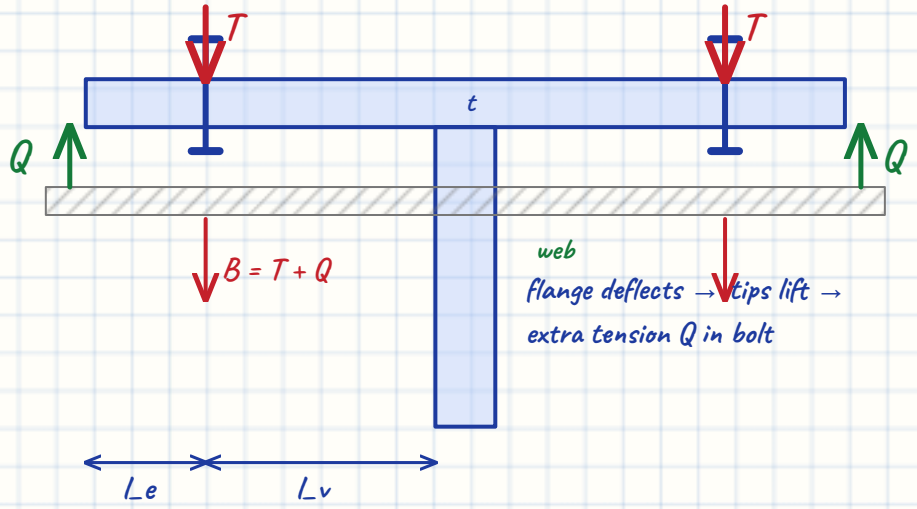
- Direct shear per bolt $V = P/n$.
- Moment $M = P\,e$ rotates the bracket about a **neutral axis**. Taking first the bottom bolts in tension:

$$T_i = M\,y_i / \Sigma\, y^2 \text{; (y measured from the point of rotation/bottom of bracket for simple approx.)}$$

- Check combined: $(V/V_{db})^2 + (T/T_{db})^2 \le 1$.
- If the end-plate is stiff and bolts are not pre-tensioned, the neutral axis is taken at the **bottom flange** (conservative) for a deep bracket; else at centroid of bolt group for short brackets.

Bolt in tension under **prying** → add Q (see next sections).

10. Prying Action



T-stub connection – prying force Q develops at flange tips

When a flexible connecting plate (T-stub, end plate, hanger) is pulled by bolts, the plate **bends**; its tips press on the support and give extra reaction **Q** → bolt force becomes **T + Q**.

IS 800 (Cl. 10.4.7) approach

$$Q = (l_v / 2l_e)\,[T_e - (\beta\,\eta\,f_o\,b_e\,t^4) / (27\,l_e\,l_v^2)]$$

- T_e = external tension per bolt, t = flange thickness, b_e = effective width of flange per bolt
- l_v = distance from bolt centre to toe of fillet weld / root of web; l_e = min(e, 1.1 $t\sqrt{(\beta f_o/f_y)}$)
- β = 2 (non-pretensioned bolts), 1 (pretensioned); η = 1.5; f_o = 0.7 f_y
- Provided: $T_e + Q \le T_{db}$ (bolt tension capacity)

Remedies / Reducing prying

- Thicker flange ($t\uparrow$) – prying $\propto 1/t^3$; bolts closer to web ($l_v\downarrow$)
- Larger edge distance, stiffeners, pre-tensioned (HSFG) bolts

> Prying is **neglected** when the flange is thick enough ($t \ge \approx 1.1\sqrt{(\dots)}$) or bolts are HSFG pre-tensioned & joint is designed as non-flexible.

11. HSFG (High Strength Friction Grip) Bolts

Made of high-strength steel (grades 8.8, 10.9), tightened to a **specified initial tension F_o = 0.7 f_{ub} A_{nb}**. Plates are clamped → load passes by **friction**, bolts are not in bearing.

Advantages

- No slip → **rigid joint**, good for fatigue, vibration, load reversal (bridges, crane girders)
- Bolt not in shear/bearing (pre-tension carries); less number of bolts than black bolts at service
- Quiet, easy to inspect with **torque/turn-of-nut** methods

Design slip resistance (IS 800 Cl. 10.4.3)

$$V_{dsf} = V_{nsf}/\gamma_{mf} ; V_{nsf} = \mu_f n_e K_h F_o$$

- μ_f = coefficient of friction (Table 20): **0.33** clean mill scale, **0.48** sand/shot-blasted, 0.10 red-lead painted...
- n_e = number of effective interfaces (friction surfaces) ; K_h = 1.0 (std. holes), 0.85 (oversize / short slot), 0.7 (long slot)
- γ_{mf} = 1.25 (1.10 if slip is serviceability limit)
- Tightening: turn-of-nut, calibrated wrench, load-indicating washers (DTI).

Tension + shear

$$V_{sf}/V_{dsf} + T_e/T_{ndf} \le 1.0 ; T_{ndf} = 0.9 F_o/\gamma_{mf}$$ (slip-resistance reduced by external tension)

EXAMPLE 4 – HSFG BOLT VALUE

M20 grade 8.8 HSFG, mill scale surfaces, std. holes, 2 friction surfaces:

A_{nb} = 245 mm² ; F_o = 0.7 × 800 × 245 = **137.2 kN**

V_{dsf} = 0.33 × 2 × 1.0 × 137.2 / 1.25 = **72.4 kN** per bolt (slip resistance).

Ultimate (bearing) check – design also for V_{db} at ULS if joint is slip-critical only at service.

12. Welded Connections – Basics

Weld = fusion of parent metal (and filler electrode) by heat – **arc welding**: shielded metal arc (SMAW), submerged arc (SAW – automatic, shop), MIG/TIG.

Advantages vs. bolts

- No holes → full section effective ; lighter (no splice plates), neat, **rigid** joints
- Quiet, can join any shape, less fabrication labour

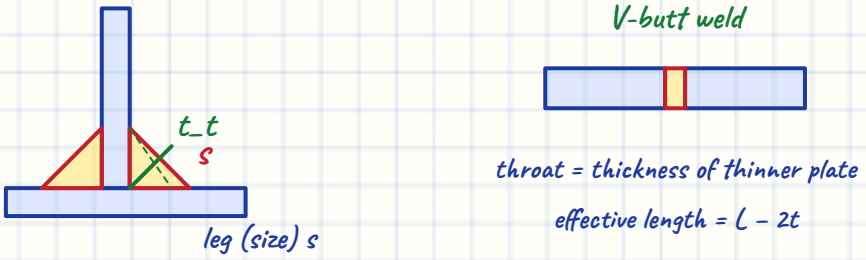
Disadvantages

- Skilled labour & inspection (NDT), **residual stresses, distortion**, fatigue crack, brittle fracture risk, difficult in field

Types of welds

- **Fillet weld** (≈ 80 % of structural welds) – triangular; used in lap & T-joints
- **Butt (groove) weld** – full or partial penetration: square, V, double-V, U, bevel
- Plug / slot weld (rare), spot weld

Joints: lap, butt, T, corner, edge.



throat $t_t = 0.7\ s$ (45° fillet)

Fillet weld – leg size s and effective throat t_t; V-butt weld

Fillet weld rules (IS 800 Cl. 10.5)

- Throat t_t = **K·s**, K = **0.7** (angle between fusion faces 60°–90°) ; 0.6 for 91°–100° ; 0.5 for 101°–106°...
- **Min. size** (by thicker part): ≤10 mm → 3 mm ; 10–20 → 5 mm ; 20–32 → 6 mm ; 32–50 → 8 mm (first run 10 mm)
- **Max. size**: edge of plate → t – 1.5 mm ; for rolled sections beyond.. ≤ thickness of thinner part
- **Min. effective length** ≥ 4 s ; Effective length = overall – 2 s
- Overlap in lap joint ≥ **4 × thickness of thinner plate** (or 40 mm)
- Longitudinal fillet welds on flat bar ends: L ≥ transverse spacing (b) ; end returns ≥ 2s
- Long welds: if $L_j > 150\ t_t$ → reduce $\beta_{lw} = 1.2 - 0.2\ L_j/(150\ t_t)$ (≥ 0.6)

13. Strength of Welds

Fillet weld (shear on throat plane)

$$f_{wd} = f_u / (\sqrt{3} \cdot \gamma_{mw}) \; (f_u = \text{smaller of weld / parent metal})$$
$$P_{dw} = f_{wd} \cdot t_t \cdot L_w \text{ (per weld)} ; \text{ strength per mm} = 0.7\,s\,f_{wd}$$

For Fe 410: shop weld $\gamma_{mw} = 1.25 \rightarrow f_{wd} = 410/(\sqrt{3}\times1.25) =$ **189.4 MPa** ; field weld $\gamma_{mw} = 1.5 \rightarrow$ **157.8 MPa**.

Strength per mm of 6 mm shop fillet = 0.7×6×189.4 = **795 N/mm**.

Butt weld

- Full penetration: strength = strength of parent metal; throat = thickness of thinner part (butt weld is "as strong as plate").
- Design stress in tension/compression = f_y/γ_{m0}, shear = $f_y/(\sqrt{3}\,\gamma_{m0})$ of weaker part.
- Incomplete penetration: throat = ≈ 5/8 × thickness, etc. – eccentricity & stress concentration.
- Effective length = L – 2t (no run-off plates)

EXAMPLE 5 – LAP JOINT WITH FILLET WELDS

Two plates 200 × 12 mm lapped, P_u = **400 kN**, shop welds, Fe 410. Choose s = 8 mm (≥ 5 mm min ; ≤ 12 – 1.5).

t_t = 0.7 × 8 = 5.6 mm ; strength/mm = 5.6 × 189.4 = 1060 N/mm

Required effective length L_w = 400 000/1060 = **377 mm**

Use two longitudinal side welds + end return : each side L_{eff} = 189 mm → overall = 189 + 2s = **205 mm** (~210 mm each side).

Check overlap ≥ 4t = 48 mm ✓ ; L ≥ b = 200 ✓ (210 mm).

14. Eccentric Welded Connections

Case 1 – In-plane eccentricity (TORSION + SHEAR) – weld group treated as a line

- Direct shear stress (per unit length): q_d = P / L_total (parallel to P)
- Torque T = P·e about centroid G of the weld group
- Torsional force per unit length at point at distance r: $q_T = T\,r / I_p$; $I_p = I_{xx} + I_{yy}$ (of weld **lines**)
- Resultant $q_R = \sqrt{[(q_d + q_{T,v})^2 + q_{T,h}^2]}$ at the critical end point.

$$\text{Need: } q_R \le f_{wd}\,t_t \text{ (per mm length)} \rightarrow \text{choose size s.}$$

Case 2 – Out-of-plane eccentricity (BENDING + SHEAR)

Bracket welded to column face by two vertical welds each L = 200 mm:

- Shear stress $q = P/(2\,L\,t_t)$; bending stress $f_b = M / Z_w$ with $Z_w = 2\,(t_t\,L^2/6) = t_t\,L^2/3$

$$f_{res} = \sqrt{(q^2 + f_b^2)} \le f_{wd}$$

EXAMPLE 6 – BRACKET ON COLUMN FACE

$P_u = 100$ kN, $e = 150$ mm, two vertical welds, $L = 200$ mm each, shop welds ($f_{wd} = 189.4$ MPa).

$q = 100\,000 / (2 \times 200 \times t_t) = 250 / t_t$

$M = 100\,000 \times 150 = 15 \times 10^6$ N·mm ; $Z_w = t_t \times 200^2/3 \rightarrow f_b = 15\times10^6 \times 3 / (40000\,t_t) = 1125 / t_t$

$f_{res} = \sqrt{(250^2 + 1125^2)}/t_t = 1152 / t_t \le 189.4 \rightarrow t_t \ge$ **6.08 mm** $\rightarrow s = 6.08/0.7 = 8.7$ mm

Provide fillet weld size 10 mm (round up ; check min size ✓).

Reminder – Always check **parent metal** (plate shear / bending) near the weld, and use $\gamma_{mw} = 1.5$ for site welding.

Quick comparison: Bolted vs Welded

Item	Bolted	Welded
Fabrication	needs holes, plates	direct, fewer parts
Site work	easy	difficult, needs power
Net area	reduced	full
Fatigue	better (HSFG)	poorer
Inspection	visual	NDT required

UNIT – II

Design of Tension and Compression Members

Syllabus: Behaviour and Design of Simple and Built-Up Members Subjected to Tension – Shear Lag Effect – Design of Lug Angles – Tension Splice – Behaviour of Short and Long Columns – Euler's Column Theory – Design of Simple and Built-Up Compression Members with Lacings and Battens – Design of Column Bases – Slab Base and Gusseted Base.

What's inside these notes

- Tension members: modes of failure, net area, block shear
- Shear lag, design of angles, lug angles, tension splice
- Column behaviour, Euler theory, effective length
- IS 800 column curves (Perry–Robertson), buckling classes
- Design of simple compression members – worked example
- Built-up columns: lacing & battens
- Column bases: slab base and gusseted base

Exam tip: Tension: write the THREE checks (yield, rupture, block shear) and take the least. Compression: find $\lambda \rightarrow f_{cd}$ from $\chi \rightarrow P_d = f_{cd} \cdot A_e$. Always check the weaker axis.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2 \times 10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Tension Members – Behaviour

Members carrying **axial tensile force** – tie rods, hangers, bottom chords of trusses, bracing, cables.

Types: rods/bars (round, flat), single angle, double angle (back-to-back), channels, I-sections, built-up (lattice) members.

Behaviour

- Under tension the member **does not buckle** → full material strength can be used.
- Bolt holes reduce the area at the connection → **net section** fails first (stress concentration at hole).
- Ductile steel: gross section yields (large elongation) before rupture ; yielding at holes redistributes stress.
- Eccentric connection (angle bolted through one leg) → **shear lag** and bending.

Design strength of tension member (IS 800 Cl. 6) – least of

(1) Yielding of gross section: $T_{dg} = A_g f_y / \gamma_{m0}$

(2) Rupture of net section: $T_{dn} = 0.9 A_n f_u / \gamma_{m1}$

(3) Block shear: T_{db} (tearing of a block at the end connection)

$\gamma_{m0} = 1.10$, $\gamma_{m1} = 1.25$. Design: $T_u \le T_d = \min(T_{dg}, T_{dn}, T_{db})$.

Slenderness limits (IS 800 Table 3)

Member	Max KL/r
Tension, normally in tension only	400
Tension – may reverse (wind / seismic)	350
Tension – reversal due to other loads	180
Compression (DL + LL)	180
Compression (wind/EQ only)	250
Compression flange of beam	300

Slenderness limit prevents sag, vibration & damage in handling.

> Rods: use $A_n = 0.78 A_g$ for threaded part (net area at threads) – $T_d = 0.9 f_u A_n/\gamma_{m1} \le f_y A_g/\gamma_{m0}$.

2. Net Area, Staggered Holes, Block Shear

Net area of plate with holes

$$A_n = (b - n d_0) t$$

Staggered holes – take the critical zig-zag line:

$$A_n = [\,b - n\,d_0 + \Sigma\, p_s^2/(4\,g)\,]\,t$$

n = holes on the failure path, p_s = stagger pitch, g = gauge between rows. The path with the **least A_n** governs.

Angles – net area

- Connected leg: $A_{nc} = (a_1 - d_0 - t/2)\,t$ (a_1 = connected leg width)
- Outstanding leg: $A_{go} = (a_2 - t/2)\,t$
- Staggered: consider the unfolded angle (gauge $g = g_1 + g_2 - t$).

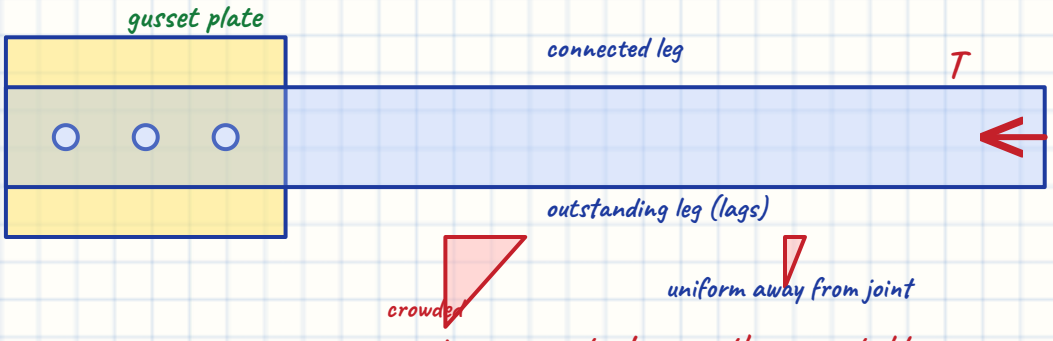
Block shear (IS 800 Cl. 6.4)

Failure by shear along bolt line(s) + tension across the end : a "block" tears out.

$$T_{db1} = A_{vg}\,f_y/(\sqrt{3}\,\gamma_{m0}) + 0.9\,A_{tn}\,f_u/\gamma_{m1}$$
$$T_{db2} = 0.9\,A_{vn}\,f_u/(\sqrt{3}\,\gamma_{m1}) + A_{tg}\,f_y/\gamma_{m0}$$
$$T_{db} = \text{smaller of the two}$$

A_{vg}, A_{vn} = gross, net area in **shear** along the bolt line ; A_{tg}, A_{tn} = gross, net area in **tension** (perpendicular to load) .

> Think: shear area **yields** while tension area **ruptures** (T_{db1}), or shear area ruptures while tension yields (T_{db2}). Use the smaller.



Shear lag in an angle connected through one leg

3. Shear Lag Effect

In angles / tees / channels connected by **only some elements** (e.g. one leg), stress is **not uniformly distributed** across the section near the joint – the connected part is overstressed, the outstanding leg "lags". Hence the **net section efficiency is reduced** and the rupture formula is modified.

Single angle with bolted/welded connection through one leg

$$T_{dn} = 0.9\,A_{nc}\,f_u/\gamma_{m1} + \beta\,A_{go}\,f_y/\gamma_{m0}$$
$$\beta = 1.4 - 0.076\,(w/t)(f_y/f_u)(b_s/L_c)\;;\; 0.7 \le \beta \le f_u\,\gamma_{m0}/(f_y\,\gamma_{m1})$$

- w = outstanding leg width ; b_s = shear lag width = $w + w_1 - t$ (bolted) ; w_1 = connected leg width
- L_c = length of end connection (outer bolt to outer bolt, or weld length)
- A_{nc} = net area of connected leg ; A_{go} = gross area of outstanding leg

For **welded**: $b_s = w$ (outstand) , L_c = weld length.

For **other sections** (channels, I): $T_{dn} = 0.9\, A_{nc}\, f_u/\gamma_{m1} + \beta\, A_{no}\, f_u/\gamma_{m1}$ … (same idea, see Cl. 6.3.3).

EXAMPLE 1 – STRENGTH OF ISA 90 × 90 × 8 IN TENSION

Angle (A_g = 1379 mm²) connected by **one leg** with **3 bolts M16 in a line, pitch 50 mm**, end distance 30 mm, edge distance 40 mm, d_0 = 18, Fe 410 (f_y = 250, f_u = 410).

1. Gross yield: $T_{dg} = 1379 \times 250 / 1.1 =$ **313.4 kN**

2. Net rupture (shear lag):

$A_{nc} = (90 - 18 - 4) \times 8 = 544$ mm² ; $A_{go} = (90 - 4) \times 8 = 688$ mm²

$w = 90$, $b_s = 90 + 90 - 8 = 172$, $L_c = 2 \times 50 = 100$ mm

$\beta = 1.4 - 0.076 \times (90/8) \times (250/410) \times (172/100) = 0.50 < 0.7 \rightarrow$ **$\beta = 0.7$**

$T_{dn} = 0.9\times544\times410/1.25 + 0.7\times688\times250/1.1 = 160.6 + 109.5 =$ **270.0 kN**

3. Block shear: $A_{vg} = 130\times8 = 1040$; $A_{vn} = (130 - 2.5\times18)\times8 = 680$; $A_{tg} = 40\times8 = 320$; $A_{tn} = (40 - 9)\times8 = 248$

$T_{db1} = 1040\times250/(\sqrt{3}\times1.1) + 0.9\times248\times410/1.25 = 136.5 + 73.2 = 209.7$ kN

$T_{db2} = 0.9\times680\times410/(\sqrt{3}\times1.25) + 320\times250/1.1 = 115.9 + 72.7 = 188.6$ kN

$T_d = \min(313.4, 270.0, 188.6) =$ **188.6 kN** (block shear governs)

4. Design of Tension Members (Procedure)

- 1. Factored tension T_u known. Assume $T_d \approx T_{dg} \rightarrow$ **A_g required = $1.1\, T_u / f_y$**; add 20-40 % for holes/shear lag (angles) → choose section from tables.
- 2. Check slenderness ≤ 350/400 using r_{min}.
- 3. Decide connection (bolts/welds); compute number & arrangement.
- 4. Check T_{dn} and T_{db} (with shear lag for angles); revise if $T_d < T_u$.
- 5. Check deflection/sag for long members (self weight) ; tack bolts for built-up.

Choosing sections

Section	Remark
Round rod	very light, bracings; threaded ends; no bending stiffness
Single angle	simple; eccentric → shear lag → reduce capacity
Double angles back-to-back	gusset between – concentric, used for truss chords/webs
Channels	good for larger loads, bolts through web
Built-up	laced / battened for long, stiff members

Built-up tension members

- Two or more components joined by **tack bolts / tack welds** at pitch ≤ **1000 mm** (tension), so the parts act together.
- Lacing / battens (like columns) – min. force 2.5 % ; slenderness of each part between tacks ≤ 50 (or 70 % of member).
- Tack bolts **do not** count for strength.

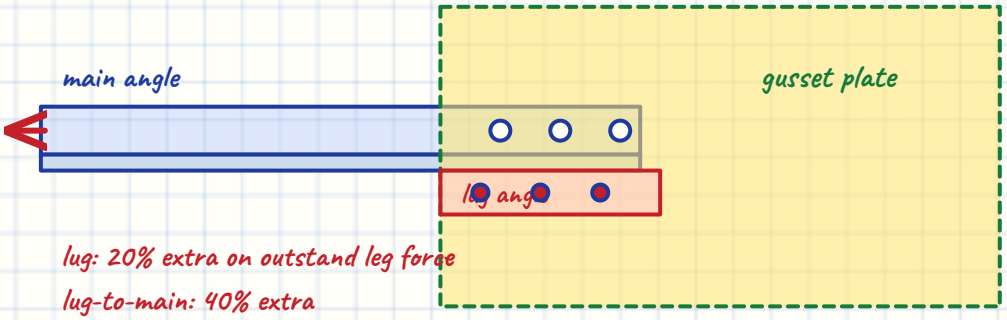
EXAMPLE 2 – REQUIRED AREA (QUICK)

T_u = 250 kN, bolted ISA pair, Fe 410.

$A_g(req) \approx 1.1 \times 250\times10^3/250 = 1100\ mm^2 \rightarrow \times1.3$ for holes/shear lag ≈ **1430 mm²**

Try **2 ISA 65x65x6** ($A = 2 \times 744 = 1488\ mm^2$) – then do three checks. Slenderness ($L = 2\ m$, $r_{min} \approx 12.6\ mm$) = 159 < 350 ✓.

5. Lug Angles



Lug angle – shortens the connection length

Problem: a long end connection in a channel/angle makes a big gusset. A **lug angle** (small angle) connects the **outstanding leg** to the gusset, so more of the section is connected → reduces length of connection & shear lag.

IS 800 requirements (Cl. 10.12)

- Lug angle + connection to gusset designed for **120 %** of the force in the outstanding leg of the main angle (20 % extra).
- Attachment of lug to main angle designed for **140 %** of that force (40 % extra).
- Effective length of the main member... lugs at least equal in thickness to the main angle; **at least 2 bolts/rivets** in the lug connection.
- Lug angles & main angle should be connected as near the end as possible.
- If lug angle is used the strength of the **member = full gross section** (no net area reduction beyond the hole line).

Lug angle design steps: Force in outstanding leg = $T \times (A_{outstand}/A_{total}) \rightarrow \times 1.2 \rightarrow$ bolts to gusset ; $\times 1.4 \rightarrow$ bolts/welds to main angle. Bolts shared by bending of the outstand.

6. Tension Splice

A splice joins two lengths of a tension member (rolled lengths limited to ~12 m).

- **Plates:** double cover plates (butt joint) – bolted or welded.
- **Angles:** cover angle (or plate) on both legs; **I-sections:** cover plates on flanges and web.

Design rule: the splice should carry the **full member strength** (or the design force) and be as strong as the member.

Steps (bolted plate splice)

- 1. Force to transmit $P_u = T_d$ of member (or actual).
- 2. Choose cover plate thickness so that $t_{cover} \geq t_{main}/2$ **each (double cover)** or total area ≥ main plate area.
- 3. Bolt value (double shear / bearing) → $n = P_u/V_d$ on each side of the joint.
- 4. Check cover plate for yield & rupture & block shear ; check length of lap.
- 5. Keep joint symmetric about the centroid → no eccentricity.

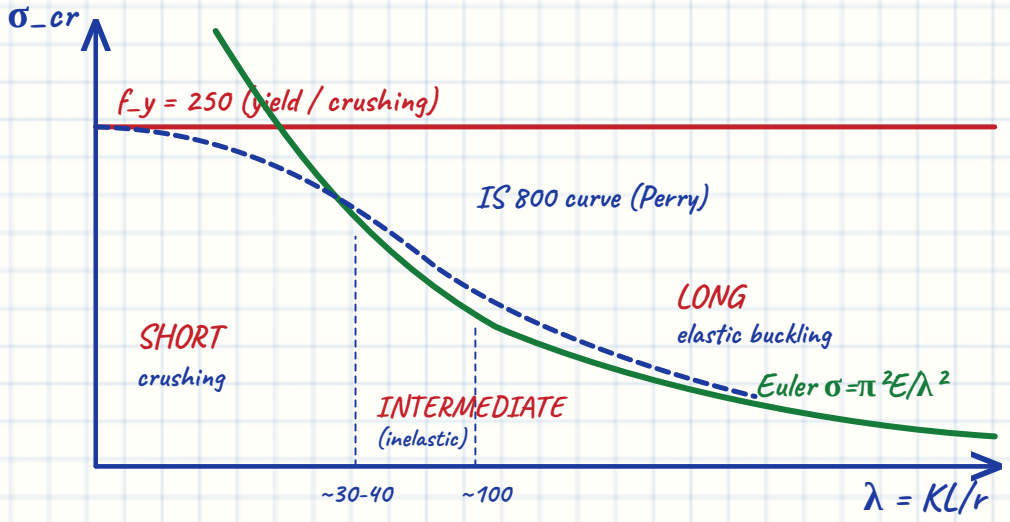
In flange splice of I-section: flange plates carry **M/d** (flange force) ; web plates carry **shear** (+ the web's share of moment).

7. Compression Members – Behaviour

Columns, struts, top chords of trusses, braces. Failure depends on **slenderness ratio** $\lambda = KL/r$ ($r = \sqrt{(I/A)}$).

- **Short column** (λ small, $\lesssim$ 30-40 for steel) – fails by **crushing/yielding** : $P = f_y A$.
- **Intermediate – inelastic buckling** (residual stresses, imperfection).
- **Long column – elastic buckling** at the Euler load (before yield).

Instability = loss of stiffness; deflection sidesway grows suddenly at P_{cr}.



Column strength vs slenderness

Factors that reduce the buckling strength

- Initial crookedness / eccentric load ; **residual stresses** (rolled & welded)
- End conditions (K) ; cross-section shape ; material non-linearity.

8. Euler's Column Theory

Assumptions: perfectly straight, centrally loaded, homogeneous, elastic (Hooke), small deflection, hinged ends, plane sections.

Derivation (hinged-hinged)

$EI\,y'' + P\,y = 0 \rightarrow y = A\sin(\pi x/L) \Rightarrow$

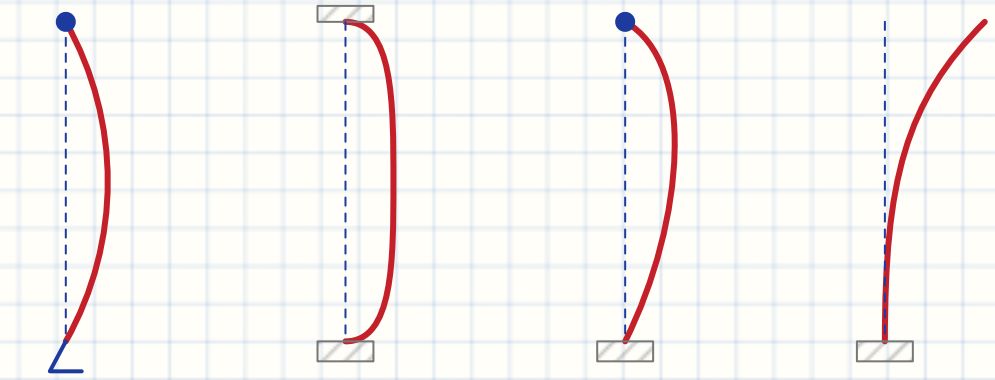
$$P_{cr} = \pi^2 E I / L^2 \text{ (buckles about weaker axis, } I_{min})$$

$$\sigma_{cr} = P_{cr}/A = \pi^2 E / \lambda^2,\ \lambda = L/r$$

With effective length $L_e = K L$

$$P_{cr} = \pi^2 E I / (K L)^2 = \pi^2 E I / L_e^2$$

Le = K · L (dotted = original, red = buckled shape)



Effective length for different end conditions

IS 800 Table 11 – effective length factors K

End conditions	K
Both ends held in position & fixed against rotation	**0.65**
Held in position at both ends, one end fixed, other hinged	**0.80**
Held in position, both ends hinged	**1.00**
One end fixed (held), other end fixed rotation but free to sway	1.20
One end fixed, other end hinged but free to sway	1.50
One end fixed, other free (cantilever)	**2.00**

Limitations of Euler's theory

- Valid only for **long columns** (λ > ≈ 100 for steel; σ_{cr} must be < proportional limit f_p).
- For $\lambda = \pi\sqrt{(E/f_y)} \approx$ **89** (f_y = 250), $\sigma_{cr} = f_y$.
- Ignores imperfections & residual stresses → real strength lower. Hence design codes use **Perry-Robertson curves** with imperfection factor α.
- Rankine formula (older): $1/P = 1/P_c + 1/P_E$.

9. Design Compressive Strength (IS 800 Cl. 7.1)

$$P_d = A_e \, f_{cd} \; ; \; f_{cd} = \chi \, f_y / \gamma_{m0} \le f_y / \gamma_{m0}$$

$$\chi = 1 / \left[\phi + \sqrt{(\phi^2 - \bar{\lambda}^2)} \right] \; ; \; \phi = 0.5 \left[1 + \alpha \, (\bar{\lambda} - 0.2) + \bar{\lambda}^2 \right]$$

$$\bar{\lambda} = \sqrt{(f_y / f_{cc})} = \sqrt{\left[f_y \, (KL/r)^2 / (\pi^2 E) \right]} \; ; \; f_{cc} = \pi^2 E / (KL/r)^2$$

α = imperfection factor ; A_e = effective area (full area for class 1-3).

Buckling class	a	b	c	d
α	0.21	0.34	0.49	0.76

Buckling class selection (Table 10, simplified)

Cross-section	Axis	Class
Rolled I ($h/b > 1.2$, $t_f \le 40$)	z-z / y-y	**a / b**
Rolled I ($h/b \le 1.2$, $t_f \le 100$)	z-z / y-y	b / c
Welded I ($t_f \le 40$)	z-z / y-y	b / c
Hot-rolled hollow (CHS, RHS)	any	**a**
Cold-formed hollow	any	b
Channels, angles, T, built-up from these	any	**c**

EXAMPLE 3 – ISMB 300 COLUMN, L = 4 M, HINGED BOTH ENDS

Properties: $A = 5626$ mm² ; $r_{yy} = 28.4$ mm (weak axis governs) ; $f_y = 250$; $E = 2 \times 10^5$.

$K = 1.0 \rightarrow \lambda = 4000/28.4 = \mathbf{140.8}$ (< 180 ✓)

$f_{cc} = \pi^2 E/\lambda^2 = 9.8696 \times 2 \times 10^5 \; / \; 140.8^2 = 99.6$ MPa

$\bar{\lambda} = \sqrt{(250/99.6)} = 1.585$

$h/b = 300/140 = 2.14 > 1.2$, $t_f = 13.1 < 40 \rightarrow$ y-y : **class b, $\alpha = 0.34$**

$\phi = 0.5[1 + 0.34(1.585 - 0.2) + 1.585^2] = 0.5[1 + 0.471 + 2.512] = 1.99$

$\chi = 1/[1.99 + \sqrt{(1.99^2 - 2.512)}] = 1/(1.99 + 1.206) = \mathbf{0.313}$

$f_{cd} = 0.313 \times 250 \; / \; 1.1 = \mathbf{71.1}$ MPa

$$P_d = 71.1 \times 5626 = 400 \text{ kN}$$

Design procedure of a simple column

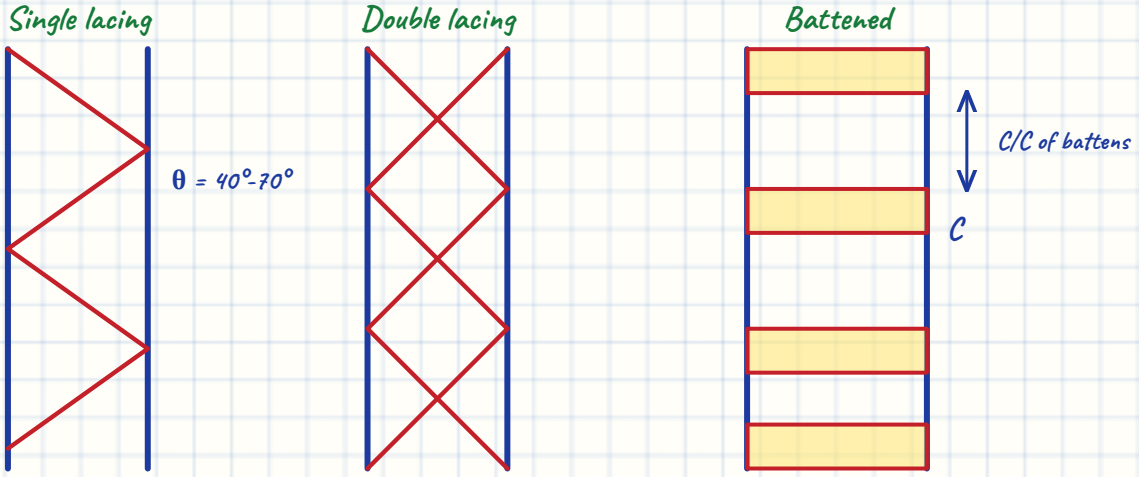
- 1. Assume $f_{cd} \approx 80$-120 MPa (for $\lambda \approx 60$-100) $\rightarrow A_{req} = P_u / f_{cd}$.
- 2. Select section (ISHB/ISMB, or built-up); get r_{min}.
- 3. Compute λ, buckling class, f_{cd} (Table 9 interpolation or formula), P_d.
- 4. If $P_d < P_u$ (or $\gg P_u$) revise. Check $\lambda \le 180$. Check local buckling (section class).
- 5. Design splice and base.

10. Built-up Compression Members

Used when rolled sections cannot give the required **area / radius of gyration** : two channels/ISMB back-to-back or toe-to-toe, box, 4 angles. Components are connected by **lacing** or **battens** (or continuous plates) so that they act as one unit.

Principle: keep $r_{zz} \approx r_{yy}$ (strong = weak axis) by spacing the components → "**effective radius**" larger.

- The **minimum distance** between channels is chosen so that $I_{yy} \geq I_{zz}$ (**equal strength about both axes**).
- Parallel axis theorem: $I_{yy} = 2\ [\ I_{own} + A\ (S/2)^2\ ]$.



Lacing (single / double) and battens

11. Lacing – Design Rules (IS 800 Cl. 7.6)

- Lacing = flats / angles / channels in zig-zag (**single**) or crossed (**double**) between the main components.
- **Angle of inclination θ** with the axis: **40° – 70°** (single), 40°–70° double.
- Slenderness of lacing bars **KL/r ≤ 145**. Effective length: L_e **= length between inner end bolts** (single, bolted) ; **0.7 × length** (welded, between welds) ; 0.7 L for double lacing too.
- Min. width of flat ≥ 3 × bolt dia ; **thickness t ≥ L_{lace}/40 (single) or /60 (double)** (L = distance between inner connections).
- **Transverse shear** V = 2.5 % of axial load in column (design lacing for V); shared equally by the lacing planes.
- Force in each lacing bar (single lacing, N planes): **F = V / (N sin θ)** ; for double lacing F = V/(2 N sin θ)
- Lacing is attached by bolts or welds; **tie plates** at ends of lacing.
- Main component slenderness between lacing connections $\lambda_1 \leq$ **50** and ≤ **0.7 × overall** λ of the member.
- **Effective slenderness ratio of laced column = 1.05 × λ_0** (λ_0 = max. actual slenderness).

Lacing: **battens at the ends** (tie plates) must be provided; lacing should not be mixed with battens in the same panel.

12. Battens – Design Rules (IS 800 Cl. 7.7)

- Battens = plates connecting main members rigidly, spaced along length – columns act like a **Vierendeel frame** (moments + shear in battens).
- Spacing C such that slenderness of **main component between battens ≤ 50** and ≤ **0.7 λ_0**.
- **Effective depth** of end batten ≥ distance between centroids of main components (a), intermediate batten ≥ 0.75 a ; **end batten ≥ 2 × width of one member**.
- Thickness t ≥ **1/50 × distance between inner edges of connections**.
- **Effective slenderness = 1.1 × λ_0** (more flexible than laced).
- At least **3 battens** per column; end ones as tie.

> Transverse shear $V = 2.5\ \%\ P_u$; longitudinal shear in batten $V_b = V\,C / (N\,S)$
> Moment in batten $M = V\,C / (2\,N)$

C = c/c spacing of battens along the length ; N = number of parallel planes of battens ; S = min. transverse distance between centroids of the connection groups.

Batten is checked for shear V_b, bending M, and its connection (bolts/welds) for shear + moment.

Procedure – design of laced/battened column

- 1. Select two channels/ISMBs for **area ≈ P_u/f_{cd}** (trial $f_{cd} \approx 100$). Check about the **minor axis** (y-y of each component) – this is a normal column check using **r_{min}**.
- 2. Choose spacing S so that **$I_{yy,total} \geq I_{zz,total}$** (usually clear spacing between toes ≥ 100 mm, ≥ 2 flange widths).
- 3. Compute effective slenderness (× 1.05 laced / × 1.1 battened) and **f_{cd}**; $P_d \geq P_u$.
- 4. Design lacing or battens ($V = 2.5\ \%\ P_u$) and connections.

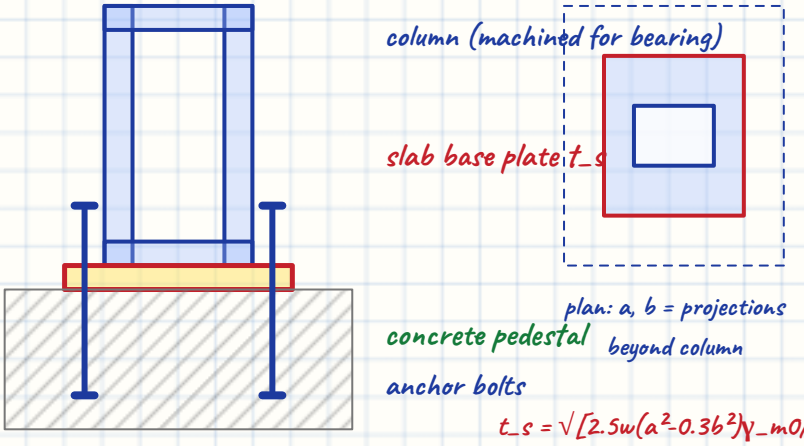
EXAMPLE 4 – LACING FORCE

Column carries P_u = 1200 kN with two lacing planes. V = 0.025 × 1200 = 30 kN → per plane = 15 kN.

Single lacing, θ = 45°: F = 15/sin 45° = **21.2 kN** in each lacing bar. Flat 50 × 6 : L_e (bolted) ≈ 250 mm ; r = 6/√12 = 1.73 mm → λ = 144 < 145 ✓ (OK, marginal) → use 50×8 for safety.

13. Column Bases – Slab Base

Column load P is spread over a large concrete area. Bearing strength of concrete (IS 456) = **0.45 f_{ck}** (confined: more).



Slab base – plate thickness from cantilever action

Design steps (IS 800 Cl. 7.4)

- 1. Required plate area $A_{req} = P_u / (0.45\, f_{ck})$. Choose B × L with projections beyond column ≥ 50–100 mm.
- 2. Bearing pressure $w = P_u/(B\,L) \le 0.45\, f_{ck}$.
- 3. Plate thickness (cantilever projection a (larger), b (smaller)):

$$t_s = \sqrt{[\,2.5\, w\, (a^2 - 0.3\, b^2)\, \gamma_{m0} / f_y\,]} \ge t_f \text{ of column flange}$$

- 4. If column end is machined for bearing, no weld needed to transmit load – **fillet weld** (tack) connection to the base plate (nominal) + **anchor bolts** (for erection, uplift, moment) ; 2–4 bolts of 20–24 mm.
- 5. Grout 25–50 mm below plate ; check shear at the base (friction + bolts).
- Plates up to **50 mm** are machined from a single slab; thicker (>50 mm) need machining of both faces.

> If bending of column exists, base has to resist **moment** → anchor bolts in tension, check bearing on one side.

EXAMPLE 5 – SLAB BASE FOR ISHB 300

P_u = 1000 kN, column ISHB 300 (h = 300, b_f = 250, t_f = 10.6), M25 concrete (f_{ck} = 25).

Permissible bearing = 0.45 × 25 = 11.25 N/mm² → A_{req} = 1000×10³/11.25 = 88 889 mm²

Provide **400 × 300** plate = 120 000 mm² → w = 1000×10³/120000 = **8.33 N/mm²** ✓

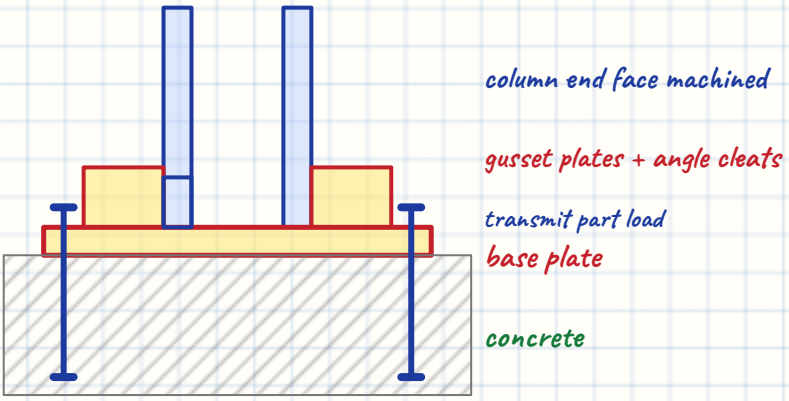
Projection: a = (400 − 300)/2 = 50 mm ; b = (300 − 250)/2 = 25 mm (choose a = larger, b = smaller)

$t_s = \sqrt{[\,2.5 \times 8.33 \times (50^2 - 0.3 \times 25^2) \times 1.1 / 250\,]} = \sqrt{(211.9)} =$ **14.6 mm** > t_f = 10.6 → provide **16 mm** plate.

Anchor bolts 4 No. M20 ; weld nominal 6 mm.

14. Gusseted Base

For **heavy loads** (large P or moments, where a plain slab would be too thick). Column end is **not** designed for bearing alone: gusset plates & cleats transfer the load.



Gusseted base

Components: column (flanges/web), **gusset plates** (on flanges), **angle cleats**, **base plate (thin, 20–30 mm)**, anchor bolts.

Design idea

- Load is shared: usually **assume 50–60 % through direct bearing** (machined column end on base plate) and rest via gussets and cleats.
- Gusset plate thickness ≥ web/flange of column, checked for bending as a cantilever carrying the portion of the load / pressure over its tributary width.
- Connection of column to gusset: bolts/welds for the load carried **by the gusset** (P × share).
- Base plate: designed as **cantilever** from face of gusset/angle, and as **continuous plate** between gussets, with $w = P/(B\ L)$:

$$M = w\,a^2/2 \text{ per unit width};\ t = \sqrt{(4\,M\gamma_{m0}/f_y)} \ldots (t^2 = 6M/(f_y/\gamma_{m0}) \text{ for rectangular section})$$

- Weld between gusset and base plate designed for the share of the load (+ moments).
- Anchor bolts for uplift (wind/moment).

Slab base vs Gusseted base

Feature	Slab base	Gusseted base
Load	light / medium	heavy
Plate thickness	thick (≥ 20 mm)	thin (gusset stiffened)
Fabrication	machined plate	welding & cleats
Cost	material high	fabrication high, lighter
Details	simple	complex

Other types : grillage footings for very heavy loads; moment-resisting bases with extended plates for portal frames.

> **Exam Q:** Always draw the base with column, base plate, grout, anchor bolts, concrete pedestal and label a, b, t, B, L.

23A01601 · DESIGN OF STEEL STRUCTURES

UNIT – III

Design of Beams

Syllabus: Design of Laterally Supported and Unsupported Beams – Design of Built-Up Beams – Design of Plate Girders.

What's inside these notes

- Beam behaviour, section classification, bending & shear capacity
- Laterally supported beam – design steps & example
- Deflection, web buckling, web crippling (bearing)
- Lateral–torsional buckling & laterally unsupported beams (example)
- Built-up beams, cover plates, curtailment
- Plate girders: proportioning, flange & web design, stiffeners
- Plate girder worked example

Exam tip: Beam design order: (1) M_u, V_u → (2) trial section → (3) classify → (4) M_d (check LTB) → (5) shear → (6) deflection → (7) web buckling / crippling at supports.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2\times10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Beams – Introduction

Beam = member carrying **transverse loads** (bending + shear). Types: simply supported, cantilever, continuous, fixed ; joists, girders (main), purlins, lintels, crane girders, floor beams.

Common sections: ISMB, ISLB, ISWB, ISMC, built-up I (plate girders), box, compound (I + plate / channel).

Failure modes of beams

- **Flexural yielding** (plastic hinge) – most economical
- **Lateral–torsional buckling (LTB)** – slender compression flange
- **Shear yielding / buckling of web**
- **Web crippling / buckling** under concentrated loads or reactions
- Local buckling of flange (slender section)
- Excess **deflection** (serviceability)

Section classification & ε (recap)

$\varepsilon = \sqrt{(250/f_y)}$. For rolled I-section (strong axis bending):

Class	Flange b/t_f (outstand)	Web d/t_w
Plastic	$\leq 9.4\ \varepsilon$	$\leq 84\ \varepsilon$
Compact	$\leq 10.5\ \varepsilon$	$\leq 105\ \varepsilon$
Semi-compact	$\leq 15.7\ \varepsilon$	$\leq 126\ \varepsilon$

Welded sections: flange 8.4ε / 9.4ε / 13.6ε.

2. Laterally Supported Beam – Bending Strength

Laterally supported = compression flange is **restrained** (slab, purlin sheeting with sufficient fixity, bracing) → no LTB; section yields.

Design bending strength (IS 800 Cl. 8.2.1.2)

$$M_d = \beta_b Z_p f_y / \gamma_{m0} \leq 1.2\, Z_e f_y / \gamma_{m0} \text{ (simply supported)}$$
$$\leq 1.5\, Z_e f_y / \gamma_{m0} \text{ (cantilever)}$$

- $\beta_b = 1.0$ for plastic & compact ; Z_e/Z_p **for semi-compact**
- Z_p = plastic section modulus ; Z_e = elastic section modulus
- Shape factor for I ≈ **1.12 – 1.15**, so limit $1.2\, Z_e f_y/\gamma_{m0}$ seldom governs.
- Slender (class 4): use effective section.

Shear strength (Cl. 8.4)

$$V_d = V_n/\gamma_{m0} ; V_n = A_v f_y/\sqrt{3}$$

$A_v = h\, t_w$ (rolled I, major axis, h = overall depth); $d\, t_w$ (welded, d = web depth).

- **Low shear** : $V \leq 0.6\, V_d$ → no reduction in M_d

- **High shear** : $V > 0.6\ V_d$ → reduced moment capacity:

$$M_{dv} = M_d - \beta\ (M_d - M_{fd}) \le 1.2\ Z_e\ f_y/\gamma_{m0}\ ;\ \beta = (2V/V_d - 1)^2$$

M_{fd} = plastic design strength of flanges alone.

> In most rolled sections with simple spans, shear is low except near short heavy loaded spans.

Design procedure (laterally supported beam)

- 1. Find factored M_u, V_u (1.5 × (DL+LL) ; include self-weight guess ≈ span/...).
- 2. Required $Z_p = M_u\ \gamma_{m0}/f_y$ (or $Z_e = Z_p$ / 1.14) → select ISMB/ISLB from tables.
- 3. **Classify** section (b/t_f, d/t_w); get M_d, check ≥ M_u.
- 4. Check **shear** $V_d \ge V_u$; low/high shear.
- 5. Check **deflection** (service load) $\delta \le$ limit.
- 6. Check **web buckling & crippling** at supports / concentrated loads.
- 7. Detail: bearing length, stiffeners if needed.

EXAMPLE 1 – SIMPLY SUPPORTED BEAM, L = 5 M, w_u = 40 KN/M (FACTORED), LATERALLY SUPPORTED

$M_u = 40 \times 5^2 / 8 =$ **125 kN·m** ; $V_u = 40 \times 5/2 =$ **100 kN**

$Z_p(req) = 125 \times 10^6 \times 1.1 / 250 = 550\ 000\ mm^3 = 550\ cm^3 \rightarrow Z_e \approx 550/1.14 = 482\ cm^3$

Try **ISMB 300** ($Z_e = 573.6\ cm^3$, $Z_p \approx 654\ cm^3$; $b_f = 140$, $t_f = 13.1$, $t_w = 7.7$, $R_1 = 14$, $I = 8603.6\ cm^4$)

Class: $b/t_f = 70/13.1 = 5.3 < 9.4$; $d = 300 - 2(13.1 + 14) = 245.8 \rightarrow d/t_w = 31.9 < 84$ → **Plastic**

$M_d = 1.0 \times 654 \times 10^3 \times 250/1.1 =$ **148.6 kN·m** ($\le 1.2\ Z_e\ f_y/\gamma_{m0} = 156.4$ ✓) > 125 ✓

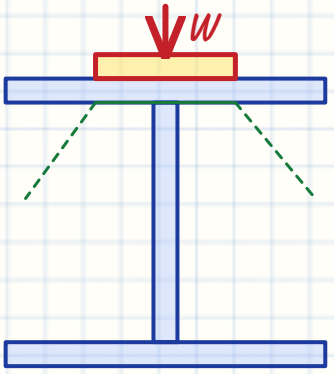
Shear: $V_d = 300 \times 7.7 \times 250/(\sqrt{3} \times 1.1) =$ **303 kN** > 100 ; $0.6\ V_d = 182 > 100$ → **low shear** ✓

Deflection (service w = 40/1.5 = 26.7 kN/m):

$\delta = 5wL^4/(384EI) = 5 \times 26.7 \times 5000^4/(384 \times 2 \times 10^5 \times 8603.6 \times 10^4) =$ **12.6 mm** < $L/300 = 16.7$ mm ✓

Use ISMB 300.

3. Web Buckling and Web Crippling (Concentrated Loads / Reactions)



1 : 2.5 dispersion through flange

→ stiff bearing length $b_1 + n_2$

web crippling $F_w = (b_1+n_2)\, t_w\, f_y/\gamma_{m0}$

web buckling $F_{cw} = (b_1+n_1)\, t_w\, f_{cd}$

(45° dispersion, $n_1 \approx d/2$ → column of web)

Dispersion of load through flange and web

Web crippling (bearing) – local yielding of web at toe

$$F_w = (b_1 + n_2)\, t_w\, f_{yw} / \gamma_{m0}$$

b_1 = stiff bearing length ; n_2 = length from dispersion at **1 : 2.5** through flange + root (to the web-flange junction).

Web buckling – web acts as a strut

$$F_{cw} = (b_1 + n_1)\, t_w\, f_{cd}$$

n_1 = dispersion at **45°** through half depth ; f_{cd} from column curve with slenderness $\lambda = 2.5\, d / t_w$ (for fixed flanges) – (effective length 0.7 d).

- If F_w or F_{cw} < reaction → provide **bearing stiffeners** or increase bearing length.
- Support reaction $R_u \le \min(F_w, F_{cw})$.

Web shear buckling

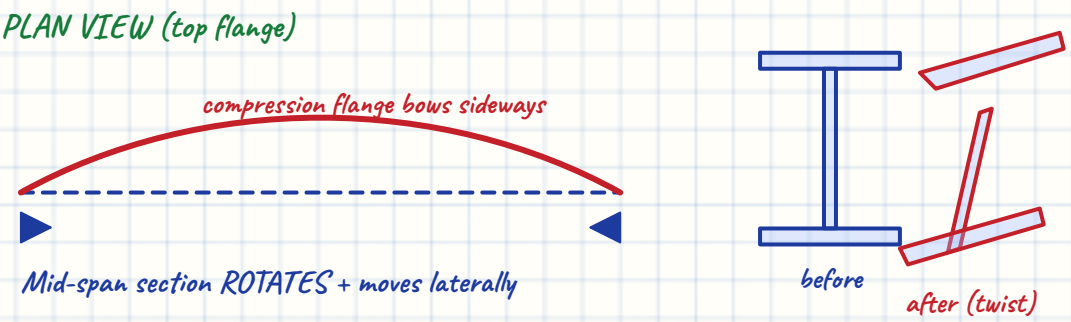
For $d/t_w \le 67\,\varepsilon$ → no check for shear buckling (rolled beams ok). For larger d/t_w : use **plate girder** shear provisions.

Deflection of beams (service load)

Case	δ_{max}
SS, UDL w	$5wL^4/384EI$
SS, central point load W	$WL^3/48EI$
Cantilever, end load W	$WL^3/3EI$
Cantilever, UDL	$wL^4/8EI$
Fixed, UDL	$wL^4/384EI$

Limits: L/300 floor beams, L/360 supporting plaster, L/150–L/180 purlins, L/750 crane girders.

4. Laterally Unsupported Beams – LTB



LTB is prevented by lateral restraint of the compression flange

Lateral–torsional buckling

When the compression flange is free to deflect sideways, the beam **buckles laterally + twists** (like a thin ruler bent flat) at a moment **less than M_p**. Strong-axis bending for slender (high I_{zz}/I_{yy}) sections.

Factors affecting LTB

- **Unbraced length L_{LT}** (↑ → ↓ strength) ; I_{yy}, torsional const. I_t, warping const. I_w
- Type of loading, **position of load** (top flange destabilising), support restraints
- Moment gradient, cross-section shape (box & tubes hardly LTB), residual stress

> LTB is **not a problem** for: box/hollow sections, bending about minor axis, beams fully restrained by slab, or if $\lambda_{LT} \le 0.4$.

Design bending strength (IS 800 Cl. 8.2.2)

$$M_d = \beta_b Z_p f_{bd} \; ; \; f_{bd} = \chi_{LT} f_y / \gamma_{m0}$$

$$\chi_{LT} = 1/[\phi_{LT} + \sqrt{(\phi_{LT}^2 - \lambda_{LT}^2)}] \le 1.0$$

$$\phi_{LT} = 0.5\,[1 + \alpha_{LT}(\lambda_{LT} - 0.2) + \lambda_{LT}^2]$$

$$\lambda_{LT} = \sqrt{(f_y / f_{cr,b})} = \sqrt{(\beta_b Z_p f_y / M_{cr})}$$

α_{LT} = **0.21** rolled sections, **0.49** welded sections.

Elastic critical moment (symmetric I-section)

$$M_{cr} = \sqrt{[(\pi^2 E I_y / L_{LT}^2)\,(G I_t + \pi^2 E I_w / L_{LT}^2)]}$$

Simplified (rolled/welded, symmetric I):

$$f_{cr,b} = [1.1\pi^2 E / (L_{LT}/r_y)^2]\sqrt{[1 + (1/20)\,((L_{LT}/r_y)/(h_f/t_f))^2]}$$

h_f = centre-to-centre of flanges ; $M_{cr} = Z_e f_{cr,b}$ … ; L_{LT} = effective unsupported length (end restraint conditions of Table 15).

Example 2 – ISMB 300 with L_{LT} = 4 m (unrestrained, simply supported)

r_y = 28.4 mm ; h_f = 300 − 13.1 = 286.9 ; h_f/t_f = 21.9 ; L_{LT}/r_y = 140.8

First term: $1.1\,\pi^2 E/(140.8)^2 = 1.1 \times 99.6 = 109.5$ MPa

Bracket : $1 + (1/20)(140.8/21.9)^2 = 1 + 2.067 = 3.067 \rightarrow \sqrt{} = 1.751$

$$f_{cr,b} = 109.5 \times 1.751 = 191.8 \text{ MPa}$$

$\lambda_{LT} = \sqrt{(250/191.8)} = \mathbf{1.142}$ (rolled $\alpha_{LT} = 0.21$)

$\varphi_{LT} = 0.5[1 + 0.21\,(1.142 - 0.2) + 1.142^2] = 0.5[1 + 0.198 + 1.304] = \mathbf{1.251}$

$\chi_{LT} = 1/[1.251 + \sqrt{(1.251^2 - 1.304)}] = 1/(1.251 + 0.511) = \mathbf{0.568}$

$f_{bd} = 0.568 \times 250/1.1 = \mathbf{129}$ MPa

$$M_d = 1.0 \times 654\times10^3 \times 129 = 84.4 \text{ kN·m (vs 148.6 kN·m when laterally supported!)}$$

So a 4 m unsupported length cuts capacity by ≈ 43 %.

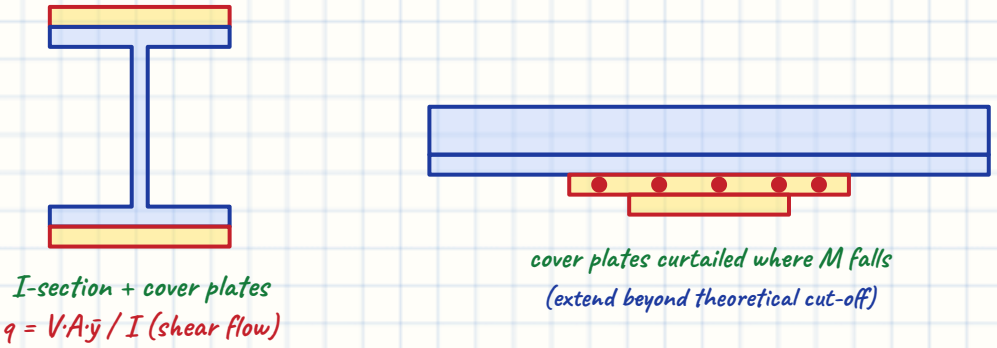
Ways to improve LTB strength

- Provide **lateral bracing / closer restraints** (cross beams, slab, diaphragms)
- Use sections with larger I_{yy} (ISWB, ISHB, box) ; add flange plate / channel on compression flange
- Lower the load point (bottom flange loading) ; ends restrained against twist

5. Built-up Beams (Compound Beams)

When a rolled section is **not enough** (heavy loads, long spans) → **strengthen** with

- **cover plates** welded/bolted to flanges ; extra **channel** on top flange ; **two I-sections** side by side ; **box** from plates ; **plate girder** (plates welded to form I).



Rolled beam with cover plates & curtailment

Design steps (I-section + cover plates)

- 1. Required Z_p ; rolled section provides part ; **extra Z from plates**: choose width b_p & thickness t_p ($b_p \le$ flange width, $t_p \le$ ~1.5 t_f total, outstand ≤ 16 t_p ε or 12 t) .
- 2. Compute I, Z_e of built-up section ; classify (cover plate outstand).

- 3. M_d, V_d, deflection, LTB as for rolled.
- 4. **Connection of cover plate to flange** – shear flow per unit length:

$$q = V A_p \bar{y} / I \ (A_p = \text{cover plate area}, \bar{y} = \text{centroid distance from N.A.})$$

- Pitch of bolts: $p = R_b / q$; or weld size from $q \le 2 \times f_{wd} t_t$ (two welds).
- 5. **Curtailment**: plate theoretical cut-off where $M_u \le M_d$ of rolled section alone. Extend by **length development** beyond cut-off: ≥ (flange width + development) so connection carries plate force. Plate end connected for its full force.
- Welds/bolts max pitch: 32 t or 300 (tension flange), 16t or 200 (compression).

EXAMPLE 3 – BOLT PITCH FOR COVER PLATE

$V = 300$ kN, $I = 25\,000$ cm^4 = 2.5×10^8 mm^4, cover plate 200 × 12 → $A = 2400$ mm², $\bar{y} = 250$ mm.

$q = 300\times10^3 \times 2400 \times 250 / 2.5\times10^8 =$ **720 N/mm**

M20 bolt single shear (4.6) ≈ 45 kN (bearing governs?) → $p = 45000/720 =$ **62 mm** → provide 60 mm c/c (≥ 2.5 d = 50 ✓, ≤ 200).

6. Plate Girders – Introduction

Large **I-section built up from plates** by welding (or bolting with angles) – used when span/loads exceed rolled sections: bridges, gantry girders, heavy floors, industrial buildings (spans 15–50 m ; depth ~ L/10 – L/12).

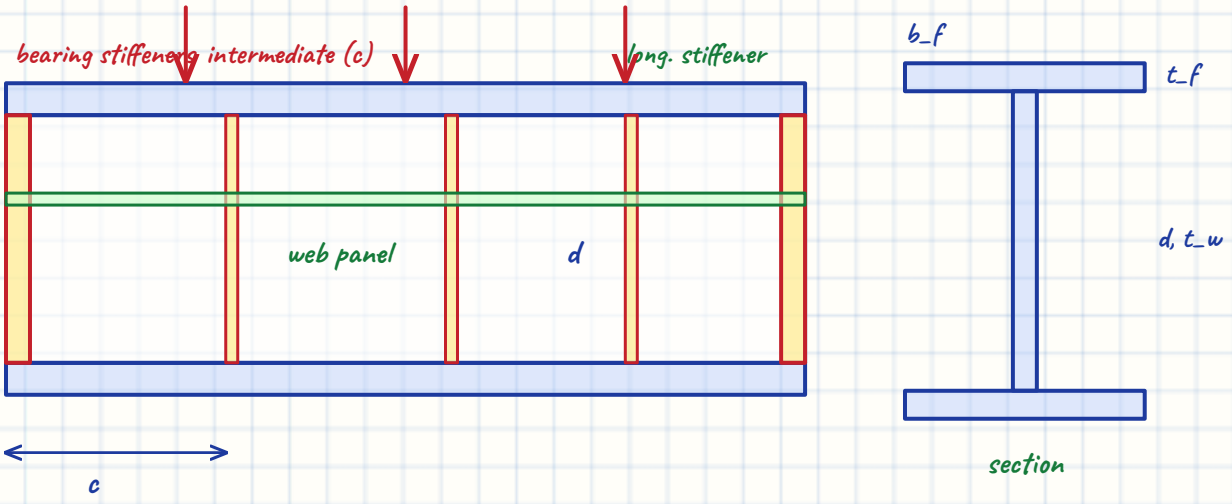


Plate girder – elevation with stiffeners and cross-section

Components: two **flanges** (carry bending: M/d), **web** (carries shear), **stiffeners** :

- **Bearing (load-carrying) stiffeners** – at supports & concentrated loads ; act as columns
- **Intermediate transverse stiffeners** – increase shear buckling resistance (spacing c)
- **Longitudinal stiffeners** – near compression flange (at 0.2 d) to reduce web slenderness
- **Flange splices, web splices**, end panels.

Advantages: material only where needed → lighter, economical, flexible depth & thickness. Disadvantages: fabrication & inspection, fatigue at stiffener welds.

7. Proportioning a Plate Girder

Depth of web d

Economical depth $d \approx (M_u / (f_y t_w))^{1/2} \cdot (...)$ – practical: **$d = L/10$ to $L/12$**. Check headroom & transport.

Web thickness t_w (IS 800 Cl. 8.6)

Serviceability (avoid flange buckling into web & excessive web distortions):

- No transverse stiffeners: **$d/t_w \leq 200\ \varepsilon$** ; with stiffeners the permissible d/t_w ↑ (up to ≈ 250-270 ε for $c < 1.5\ d$, and **$345\ \varepsilon^2$** limit to prevent flange buckling into the web). Choose $t_w \geq 8$ mm (min 6 mm – exposed).

Flange area A_f

Flanges carry the moment, web assumed to carry shear only (**flange area method**):

$$A_f \approx M_u \gamma_{m0} / (f_y\ d_f)\ ;\ d_f = \text{distance between flange centroids} \approx d + t_f$$

Adjust for web contribution: $A_f = M/(f_y\ d) - A_w/6$ (approx.).

- Flange outstand $b/t_f \leq$ **13.6 ε** (semi-compact) / 8.4ε (plastic) ; $b_f \approx d/5$ to $d/3$.
- Flange plate thickness, taper/curtail in 1–2 steps (flange plate changes) where M reduces.

8. Plate Girder – Strength Checks

(a) Bending (flange)

$$M_d = A_f\ d_f\ f_y / \gamma_{m0}\ (+ \text{web contribution for plastic design}) \geq M_u\ ;\ \text{for LTB use } \chi_{LT}\ (\alpha_{LT} = 0.49)$$

(b) Shear strength of web (IS 800 Cl. 8.4.2 – simple post-critical method)

$$V_d = V_{cr}/\gamma_{m0}\ ;\ V_{cr} = A_v\ \tau_b = d\ t_w\ \tau_b$$

Elastic critical shear stress:

$$\tau_{cr,e} = k_v\ \pi^2 E / [12\ (1 - \mu^2)\ (d/t_w)^2]$$

$k_v = 5.35$ (no intermediate stiffeners) ; $4 + 5.35/(c/d)^2$ for $c/d < 1$; $5.35 + 4/(c/d)^2$ for $c/d \geq 1$

$$\lambda_w = \sqrt{(f_{yw} / (\sqrt{3}\ \tau_{cr,e}))}$$

λ_w	τ_b
≤ 0.8	$f_{yw}/\sqrt{3}$
$0.8 - 1.2$	$[1 - 0.8(\lambda_w - 0.8)]\, f_{yw}/\sqrt{3}$
≥ 1.2	$f_{yw}/(\sqrt{3}\,\lambda_w^2)$ $(= \tau_{cr,e})$

Tension-field method (Cl. 8.4.2.2) gives higher V_d (needs end panel + stiffeners anchoring).

(c) Interaction M–V

If $V > 0.6\, V_d$: M reduced (as beams) ; for web carrying part of moment.

(d) Stiffeners

- **Intermediate stiffener**: min $I_s \ge 0.75\, d\, t_w^3$ $(c/d \ge \sqrt{2})$; $\ge 1.5\, d^3 t_w^3/c^2$ $(c/d < \sqrt{2})$; outstand $\le 12\, t_s\, \varepsilon$; spacing $c \le 3d$ (1.5 d typical).
- **Bearing stiffener** : act as a **column** of cross-section = stiffener plates + 20 t_w web strip each side ; effective length 0.7 d ; $F_x \le P_d$; also bearing check on the stiffener contact area (ground ends) : $F_{psd} = A_p\, f_y/\gamma_{m0} \ge R_u$. Weld to web for full reaction.
- **Web–flange weld** : $q = V\, A_f\, \bar{y}\, /\, I$ → weld size (two fillet lines).

EXAMPLE 4 – SIMPLY SUPPORTED PLATE GIRDER, L = 20 M

w_u = 50 kN/m (including self-weight) → $M_u = 50 \times 20^2/8 =$ **2500 kN·m** ; $V_u =$ **500 kN**.

Depth $d = L/11 \approx$ **1800 mm** ; $t_w = 10$ mm → $d/t_w = 180 < 200\, \varepsilon$ $(=200)$ ✓

Flange area $A_f = 2500\times10^6 \times 1.1 / (250 \times 1810) \approx$ **6080 mm²** → provide **400 × 16** ($A = 6400$ mm²) ; outstand $= (400 - 10)/2 = 195 \to 195/16 = 12.2 < 13.6$ ✓

Shear : stiffener spacing $c = 1800$ $(c/d = 1)$ → $k_v = 5.35 + 4/1 =$ **9.35**

$\tau_{cr,e} = 9.35 \times \pi^2 \times 2\times10^5 / [12 \times 0.91 \times 180^2] =$ **52.2 MPa**

$\lambda_w = \sqrt{(250/(\sqrt{3} \times 52.2))} =$ **1.66 ≥ 1.2** → $\tau_b = \tau_{cr,e} = 52.2$

$V_{cr} = 1800 \times 10 \times 52.2 = 939$ kN → $V_d = 939/1.1 =$ **854 kN > 500** ✓

Web–flange weld: $q = V\, A_f\, \bar{y}/I$ → $I \approx 2\times6400\times905^2 + 10\times1800^3/12 = 1.05\times10^{10} + 4.9\times10^9 = 1.5\times10^{10}$ mm⁴ ; $\bar{y} \approx 905$ → $q = 500\times10^3\times6400\times905/1.5\times10^{10} = 193$ N/mm → two fillet welds: 96.5 N/mm each → **s = 3–4 mm → min size 5 mm.**

Bearing stiffener at support: $R_u = 500$ kN → 2 plates 110 × 12 + web strip 2×(20×10) …

Order for exam answer: loads → depth → web thickness → flange → check bending → shear (k_v, λ_w, τ_b) → stiffeners (intermediate, bearing) → welds → splices & sketch.

9. More on Laterally Unsupported Beams

When can a beam be treated as laterally supported? (IS 800 Cl. 8.2.2)

- Compression flange **fully restrained** against lateral deflection & rotation (e.g. cast in concrete slab / connected to slab with shear connectors, sheeting with positive connection)
- Hollow (box, tube) sections and I-sections bending about the **minor axis**
- Closely spaced **bracings** so that $\lambda_{LT} \le 0.4$ (i.e. L_{LT}/r_y small)

- Slab bearing on top flange **with friction only is NOT enough** – need positive connection.

Effective length L_{LT} for LTB (Table 15 – selected)

Restraint at supports	L_{LT}
Compression flange fully restrained laterally; ends torsionally restrained (simply supported beam)	1.0 L
Both flanges free to rotate on plan (fork support), no intermediate restraints	1.2 L
Partial torsional restraint at ends	1.2 L – 1.0 L
Cantilever – fixed end restrained, free end unrestrained	1.0 L (load at tip: 1.4–2.5 L)
Load destabilising (acts at top flange, free to move)	add 20 % (×1.2)

L = span or distance between lateral restraints.

Procedure for design of unsupported beam (trial & error)

- 1. Find M_u. Assume $f_{bd} \approx 100$–130 MPa → $Z_p(req) = M_u/f_{bd}$ → select section.
- 2. Compute L_{LT}/r_y, h_f/t_f → $f_{cr,b}$ → λ_{LT} → φ_{LT} → χ_{LT} → f_{bd}.
- 3. $M_d = \beta_b Z_p f_{bd} \geq M_u$. If not, choose a deeper/wider flange section or reduce L_{LT} by bracing.
- 4. Check shear, deflection, web crippling exactly as laterally supported beam.

> Quick check: ISMB 300 (L=4 m) → $M_d \approx 84$ kN·m ; if bracing at mid-span reduces L_{LT} to 2 m → f_{bd} rises (λ_{LT} falls to ≈ 0.75) ⇒ $M_d \approx 120$ kN·m. **Bracing is cheaper than a bigger section!**

EXAMPLE 5 – HIGH SHEAR CHECK

Simply supported beam ISMB 400, span 3 m, central factored point load 400 kN (laterally supported). Properties: h = 400, t_w = 8.9 mm, $Z_p \approx 1176$ cm³ , Z_e = 1022.9 cm³, b_f = 140, t_f = 16.0.

$M_u = 400 \times 3/4 =$ **300 kN·m** ; $V_u =$ **200 kN**.

$V_d = 400 \times 8.9 \times 250/(\sqrt{3} \times 1.1) =$ **467 kN** ; $0.6\ V_d = 280 > 200$ → low shear.

$M_d = 1176 \times 10^3 \times 250/1.1 =$ **267 kN·m** < 300 × ⇒ section fails in bending. Try ISMB 450 ($Z_p \approx 1533$ cm³ → M_d = 348 kN·m) ✓.

If V were 300 kN (> 0.6 V_d) → $\beta = (2 \times 300/467 - 1)^2 = 0.081$ → $M_{dv} = M_d - 0.081 (M_d - M_{fd})$ – small reduction.

10. Bearing Stiffeners & End Bearing of Beams

Where a heavy reaction / concentrated load acts and web crippling / buckling check fails → provide **load-carrying stiffeners** (plates welded each side of the web, fitted tightly to the flange).

Design of a load-bearing stiffener

- 1. Required **bearing area** : $A_{req} = R_u \gamma_{m0} / f_y$ (contact area of stiffener ends with the loaded flange, excluding corner snips).
- 2. Choose stiffener plates (2 No.) : outstand $b_s \le$ **12 $t_s \varepsilon$** ; thickness $t_s \ge b_s/12\varepsilon$... (usually ≥ 8–10 mm)
- 3. **Column check**: effective section = 2 stiffeners + web strip of width **20 t_w** on each side (or 25 t_w for end stiffener) ; effective length = **0.7 d** (both ends fixed) → r, λ, f_{cd} → $P_d = f_{cd} A_{eff} \ge R_u$.
- 4. **Weld** stiffener to web : length d, force R_u → fillet weld length 2(d − 2×15) ... size from $q = R_u/(\Sigma L_w)$.

EXAMPLE 6 – BEARING STIFFENER AT SUPPORT OF A PLATE GIRDER

R_u = 500 kN ; web 1800 × 10, flange 400 × 16 ; f_y = 250. Take 2 plates 120 × 12, ground to bear.

(i) Bearing area (excluding snip 15 mm) : 2 × (120 − 15) × 12 = 2520 mm² → capacity = 2520 × 250/1.1 = **573 kN** > 500 ✓

(ii) Column : A_{eff} = 2 × 12 × 120 + (2 × 20 × 10 + 10) × 10 = 2880 + 4100 = **6980 mm²**

$I_{web\ axis} \approx 12 \times (2 \times 120 + 10)^3/12 = 12 \times 250^3/12 = 1.56\times10^7$ mm⁴ → $r = \sqrt{(1.56\times10^7/6980)} = 47.3$ mm

$KL/r = 0.7 \times 1800/47.3 =$ **26.6** → $f_{cd} \approx$ **195 MPa** (class c) ⇒ $P_d \approx 195 \times 6980 =$ **1360 kN** > 500 ✓

(iii) Weld to web : force 500 kN over 2 × 1700 mm length → 147 N/mm → minimum 5 mm fillet ✓.

Web splices & flange splices

- **Flange splice** : cover plates with area ≥ flange area; bolts/welds for flange force.
- **Web splice** : plates both sides ; carries shear + moment share $M_w = M \times (I_w/I)$.
- Locate splices where moments are low ; full penetration butt welds are best (shop).

11. Plate Girder – Design Steps (Complete Flow)

- 1. **Data** : span L, loads (UDL, concentrated), f_y ; reactions R_u, M_u, V_u (include self-weight ≈ 2-3 kN/m initial).
- 2. **Web depth** d = L/10 to L/12 (≥ economical depth) ; **web thickness** t_w from $d/t_w \le 200\varepsilon$ (no stiffener) or from shear; minimum 8 mm.
- 3. **Flanges** : $A_f = M_u \gamma_{m0}/(f_y\, d)$; choose b_f & t_f ($b_f \approx 0.3$-$0.4\ d$; $t_f \ge b_f/(2\times13.6\varepsilon)$) ; extra **cover plate** where M is high (curtail).
- 4. **Moment of inertia & Z_e** ; classify ; $M_d \ge M_u$ (check LTB if L_{LT} large).
- 5. **Web stiffener spacing** c : choose c ≈ d ... 1.5 d (end panel c ≤ d) ; shear V_d from simple post-critical method (or tension field) ≥ V_u at each panel.
- 6. **Shear-moment interaction** in panels with high V and M.

- 7. **Bearing stiffeners** at supports & point loads ; **intermediate stiffeners** $I \geq$ limit ; connect with welds.
- 8. **Web–flange connection** weld : $q = V A_f \bar{y} / I$.
- 9. **Splices**, deflection check ($\delta = 5wL^4/384EI$ with service loads $\leq L/300$ or $L/360$).
- 10. Draw elevation & cross-sections: label d, t_w, b_f, t_f, c, stiffener sizes, weld sizes.

Tension field action (idea only)

After the web buckles in shear (**post-buckling**), it can still carry additional shear as **diagonal tension** between stiffeners; flanges & stiffeners act like a **Pratt truss** (diagonal tension band + stiffeners strut). Gives ≈ 20–50 % more shear. Needs strong end panels & anchorage.

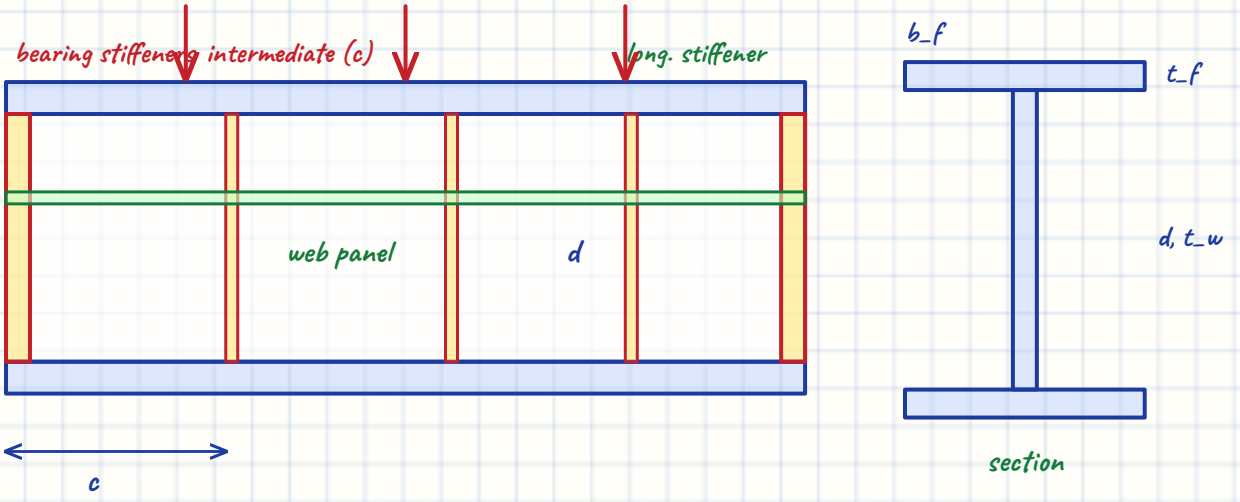


Plate girder – shear resisted by web, moment by flanges

12. Rolled Beam vs Built-up vs Plate Girder

Feature	Rolled beam	Built-up (cover plates)	Plate girder
Span range	up to ~ 10-12 m	~12-20 m	15 – 50+ m
Depth	≤ 600 mm (ISMB 600)	600-900	1–3 m
Fabrication	none	simple	complex
Design basis	table M_d	M_d + connection of plate	flange + web + stiffener design
Typical use	floors, purlins	heavy floors	bridges, gantry, heavy industrial

Important formulae – Unit III (quick sheet)

$$M_d = \beta_b Z_p f_y/\gamma_{m0} \le 1.2 Z_e f_y/\gamma_{m0}$$

$$V_d = h t_w f_y/(\sqrt{3}\gamma_{m0})\ ;\ \text{low shear if } V \le 0.6 V_d$$

$$\delta = 5wL^4/(384 E I)$$

$$f_{bd} = \chi_{LT} f_y/\gamma_{m0}\ ;\ M_d = \beta_b Z_p f_{bd}$$

$$\tau_{cr,e} = k_v \pi^2 E/[12(1-\mu^2)(d/t_w)^2]\ ;\ V_{cr} = d t_w \tau_b$$

$$q = V A \bar{y} / I \text{ (shear flow)}\ ;\ F_w = (b_1 + n_2) t_w f_y/\gamma_{m0}$$

Important exam questions

- 1. Differentiate laterally supported and unsupported beams. Explain LTB and factors.
- 2. Design a simply supported beam (L = 6 m, w = 25 kN/m DL+LL) with full lateral support.
- 3. Design a beam with L_{LT} = 3 m – check χ_{LT}.
- 4. Design a welded plate girder for span 18 m : proportioning, stiffeners, weld.
- 5. Explain web buckling, web crippling and design of bearing stiffeners.
- 6. Design cover plates for I-beam and their bolted connection ; curtailment.

UNIT – IV

Industrial Structures

Syllabus: Design of Roof Trusses – Loads on Trusses – Purlin Design Using Angle and Channel Sections – Truss Design, Design of Joints and End Bearings – Design of Gantry Girder – Introduction to Pre-Engineered Buildings.

What's inside these notes

- Industrial buildings & roof truss types
- Loads on trusses (DL, LL, wind IS 875) & combinations
- Purlin design – angle & channel (worked example)
- Truss analysis & member design, joints, gusset plates
- End bearings (fixed & roller)
- Gantry girder: loads, section, checks, example
- Pre-Engineered Buildings (PEB)

Exam tip: Truss answers need a neat sketch with panel points, loads on joints, a table of member forces (+ tension / – compression) and then member + joint design.

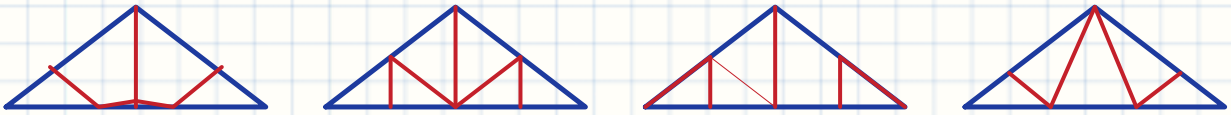
Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take $E = 2\times10^5$ MPa, $f_y = 250$ MPa, $f_u = 410$ MPa unless stated.

1. Industrial Structures – Introduction

Single-storey **industrial buildings** – factories, workshops, warehouses, power houses, hangars. Main parts:

- **Roof covering** (GI / AC / colour-coated sheets) → **purlins** → **roof trusses** → columns / stanchions → foundations
- **Bracings** (vertical, horizontal – wind, stability), **girts / sheeting rails** for walls, **gantry girders** with crane
- **Bay** = spacing of trusses (4–6 m economical) ; span 10 – 40 m.

Roof truss types



Common roof truss shapes

- **Fink / Fan** – small to medium spans ; short web members in compression
- **Howe** – diagonals in compression, verticals tension ; **Pratt** – diagonals tension (long members in tension → economic)
- **North-light**, **Mansard**, **Warren**, **Saw-tooth**, **Bowstring / parallel-chord** (flat roofs)
- Rise : **1/5 – 1/4 span** typical ; **slope ≥ 15°** for AC sheet roofs (or 1 in 3 for tiles) ; Pratt/Howe for 6–15 m span.
- Panel length chosen so purlins fall at **panel points** (no bending in rafters) ~ 1.5–2.5 m.

Sheeting

Sheet	Weight (approx.)	Remarks
GI corrugated	0.13 – 0.15 kN/m²	light, quick
AC sheet	0.17 kN/m²	brittle, min. slope 15°
Colour-coated profiled	0.05–0.10 kN/m²	PEB, modern

2. Loads on Roof Trusses

(a) Dead loads (DL)

Weight of sheeting + fixtures (0.15–0.25 kN/m²), purlins (≈ 0.10 kN/m²), **truss self-weight ≈ (L/3 + 5) × 10 N/m² of plan area** (L = span in m ≈ 0.12–0.20 kN/m²), bracings.

(b) Imposed (live) load – IS 875 (Part 2)

- Roof **not accessible** except for maintenance : **0.75 kN/m²** (slope ≤ 10°) ; reduce by **0.02 kN/m² per degree** above 10° (min. **0.4 kN/m²**), measured on **plan area**.

- Plus 0.9 kN concentrated at the mid-span of purlins (check) ; dust/snow loads where relevant.

(c) Wind load – IS 875 (Part 3)

$$V_z = V_b\, k_1\, k_2\, k_3\, k_4 \text{ (m/s)}$$

$$p_z = 0.6\, V_z^2 \text{ (N/m}^2\text{)} ; \; p_d = K_d\, K_a\, K_c\, p_z$$

$$F = (C_{pe} - C_{pi})\, A\, p_d \text{ (normal to surface)}$$

- V_b = basic wind speed (zone) ; k_1 risk coeff ; k_2 terrain & height ; k_3 topography ; k_4 importance (cyclonic).
- C_{pe} external pressure coefficients (slope, h/w ratio, windward/leeward) ; C_{pi} internal : **±0.2** (low permeability), **±0.5** (medium), **±0.7** (large openings).
- Critical case for light roofs: **suction (uplift)** may exceed DL → **net uplift** governs purlin & fastener design & stability (0.9 DL + 1.5 WL).

(d) Other: seismic (light roofs - small), snow, crane (columns), temperature.

Load combinations (strength)

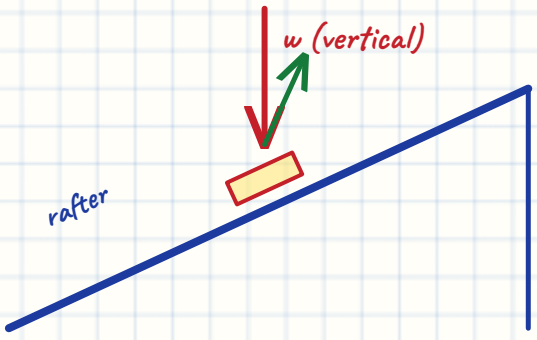
$$1.5\,(DL + LL) ; \; 1.2\,(DL + LL + WL) ; \; 1.5\,(DL + WL) ; \; 0.9\,DL + 1.5\,WL$$

Wind is applied **normal to the sloping roof surface**; resolved into joint loads : P = p × (purlin spacing along slope) × (truss spacing).

> Panel-point loads : W_{joint} = w (kN/m²) × (c/c truss spacing) × (c/c purlin spacing on plan). Apply **half loads** at eave/ridge joints.

3. Design of Purlins

Purlins = beams spanning between trusses, carrying sheeting load. Sections : **angle (ISA)** (small spans ≤ 4 m), **channel (ISMC)**, **I-section / tubular / cold-formed Z / C** (larger spans).



θ = roof slope

$w_n = w \cos\theta$ (⊥ roof) → M_x

$w_p = w \sin\theta$ (∥ roof) → M_y

sag rods at L/3 (or L/2) reduce M_y

both axes: $M_x/M_{dx} + M_y/M_{dy} \le 1$

span = truss spacing

Purlin on an inclined rafter – load components

Load on purlin (per metre length)

- Spacing along slope s ; $w = (DL_{sheet} + DL_{purlin}) + LL$ (plan → × cosθ)… ; factored $w_u = 1.5\,w$ (or wind combination, **w_n only**).
- Component **normal to roof** : $w_n = w\cos\theta$ → bending about **major axis** (M_x)
- Component **parallel to roof** : $w_p = w\sin\theta$ → bending about minor axis (M_y) (reduced by **sag rods**).

$$M_x = w_n L^2/10 \text{ (continuous / approx.) or } w_n L^2/8 \text{ (SS)} ; M_y = w_p (L/n)^2/10 \ (n = \text{no. of sag-rod spaces})$$

- **Wind load** acts ⊥ roof (no w_p) — check **uplift** with DL (0.9DL + 1.5WL).

Design steps

- 1. Compute w_n, w_p (factored). 2. M_x, M_y, shear.
- 3. Trial: $Z_{p,req} = M_x \gamma_{m0}/f_y$ (×1.2–1.4 for biaxial) → choose ISMC / ISA (also with deflection: **$\delta \le L/150$ (elastic cladding)**).
- 4. **Biaxial check**: $M_x/M_{dx} + M_y/M_{dy} \le 1.0$ (or $M_y/(Z_{py} f_y/\gamma_{m0})$ …) ; $M_{dy} \le 1.2\, Z_{ey} f_y/\gamma_{m0}$.
- 5. Shear (low), deflection (separately for both axes: $\delta = \sqrt{(\delta_x^2 + \delta_y^2)} \le L/150$).
- 6. Check **web crippling** at the bearing and **sag rod** (tension = $w_p L \times k$; $\varphi \ge 12$ mm).
- 7. Check **LTB** – roof sheeting prevents it if fixed to top flange (assume supported).

Why sag rods? w_p acts along slope; channel/angle is weak about minor axis → rods at L/3 (or mid) pick up w_p and pass it to the **ridge purlin / tie** . Sag rod diameter ≥ 12 mm.

Example 1 – Purlin loads and design moments

Truss spacing $L = 4$ m ; purlin spacing along slope $s = 1.5$ m ; roof slope $\theta = 20°$; AC sheets.

DL : sheet + fixtures $0.25 \text{ kN/m}^2 \times 1.5 = 0.375$ kN/m ; purlin self-wt 0.07 kN/m → **0.445 kN/m** (on slope; per m run)

LL (plan) : $0.75 - 0.02\,(20 - 10) =$ **0.55 kN/m²** → on purlin = $0.55 \times 1.5 \times \cos 20° = 0.775$ kN/m (vertical)

Total vertical $w = 0.445 + 0.775 =$ **1.22 kN/m** ; factored $w_u = 1.5 \times 1.22 =$ **1.83 kN/m**

$w_n = 1.83\cos 20° =$ **1.72 kN/m** ; $w_p = 1.83\sin 20° =$ **0.63 kN/m**

$M_x = 1.72 \times 4^2/10 =$ **2.75 kN·m** (continuous) ; with 1 sag rod at mid-span : $M_y = 0.63 \times 2^2/10 =$ **0.25 kN·m**

Required $Z_{p,x} \approx 2.75\times10^6 \times 1.1/250 = 12.1$ cm³ ; with biaxial factor ×1.4 ≈ 17 cm³ → very light – deflection governs : $\delta = 5 w_n L^4/(384EI) \le L/150 = 26.7$ mm → $I \ge 5 \times (1.72/1.5 = 1.15) \times 4000^4/(384 \times 2\times10^5 \times 26.7) = 7.2\times10^5$ mm⁴ = 72 cm⁴ → **ISMC 75** ($I_{xx} \approx 76$ cm⁴) or ISA 90×90×6 → check biaxial & deflection with exact table values.

4. Analysis and Design of Truss Members

Analysis

- Assume **pin-jointed truss**, loads only at joints. Reactions by statics (check uplift for wind).
- **Method of joints** or **method of sections** or graphical (Maxwell–Bow); tabulate member forces for **DL, LL, WL (left/right)** ; take critical combination ; (+ tension, − compression). Reversal in members possible under wind → design for both.
- Secondary bending ignored if members are concentric at joints.

Members & minimum sizes

Member	Section	Notes
Top chord (rafter)	**double angle** (back to back) or tee	compression ; purlins at panel points ; r_{min} governs
Bottom chord	double angle	tension; net area with holes
Web struts	single/double angle	long struts: compression
Web ties	single angle / rods	tension

Min. angle size **ISA 50 × 50 × 6** (thickness ≥ 6 mm exposed ; 8 mm if main) ; slenderness limits as Unit II (180 compression, 350 reversing, 400 tension).

Effective length of compression members in trusses (IS 800 Cl. 7.5)

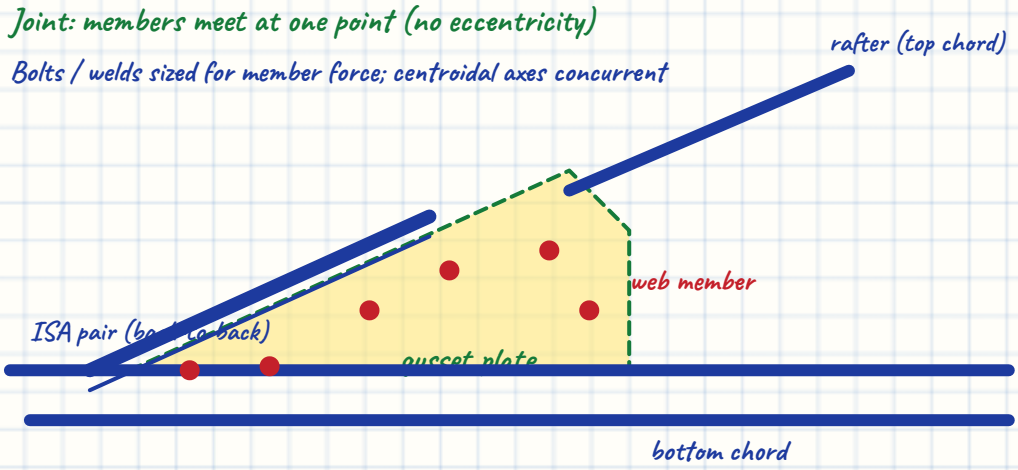
Member	Axis	Le/L (bolted / welded)
Chord – in-plane (v–v, between joints)	v–v	0.85 L
Chord – out-of-plane	y–y	1.0 L (distance between lateral supports, purlins/bracing)
Web (single angle)		0.7 L – 1.0 L

Two angles (back-to-back, tack-bolted) behave as a **T-section** — tack bolts at ≤ 600 mm. Gusset thickness 8–12 mm (by largest force).

Design of a compression chord (procedure)

- Assume $f_{cd} \approx 100$ MPa → $A_{req} = P_u/f_{cd}$; select 2 ISA ; (r_{xx} for in-plane, r_{yy} with gusset spacing for out-of-plane).
- $\lambda = \max(0.85\,L/r_{xx},\ L_y/r_{yy})$; **buckling class c** → f_{cd} → $P_d \geq P_u$. For double angles connected by tack bolts, check the slenderness of a single angle between tack bolts (≤ 0.7 × overall λ, tack bolt pitch ≤ 600 mm).
- **Tension member** (bottom chord): T_{dg}, T_{dn} (shear lag), T_{db} as per Unit II.

5. Design of Joints & End Bearings



Typical truss joint – members meeting at gusset plate

Joints

- Members joined with **gusset plate** (8–12 mm), bolts (M16–M20, 4.6/8.8) or fillet welds.
- **Centroidal axes of members meet at one point** (avoid eccentricity). For double angles, gusset passes between.
- **Number of bolts** for each member : n = Force/(bolt value) ; **bolt value** = min(shear (single/double), bearing on gusset or angle) ; or **weld length** = Force/(0.7 s f_{wd}) split between two angle legs (inversely with distances from centroid).

- Weld on angle legs: $L_1 = P\ y_2/(y_1+y_2)/(\text{strength})$, $L_2 = P\ y_1/(y_1+y_2)/(\text{strength})$

- Check **block shear** on gusset & angles; check gusset plate for tearing.
- Heel joint (rafter + bottom chord at support): gusset + bearing plate; **end bearing stiffener**.
- Ridge joint : both rafters connected via gusset or splice plate ; provide web members at purlin nodes.
- **Splice** of chords : cover plates for full force; staggered.

EXAMPLE 2 – WELDED JOINT OF ISA 75×75×8 TIE TO GUSSET

Force T_u = 150 kN, shop weld s = 6 mm (f_{wd} = 189.4) → strength = 0.7×6×189.4 = **795 N/mm**.

Required L_w = 150 000/795 = 189 mm. Distance of centroid from back: 21.4 mm → y_1 = 21.4, y_2 = 75 − 21.4 = 53.6.

L_1 (nearer heel) = 189 × 53.6/75 = **135 mm** ; L_2 = 189 × 21.4/75 = **54 mm** → plus 2 × 6 mm end returns. Provide L_1 = 140+, L_2 = 60 mm.

End bearings (supports of truss)

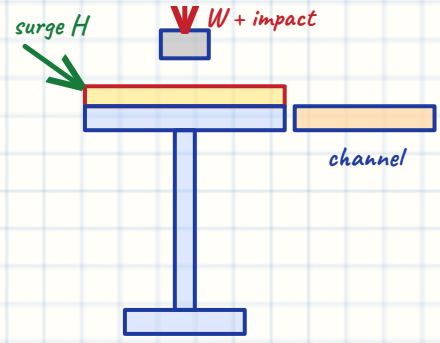
- **Fixed end** (one support) : base plate on masonry/concrete with **anchor bolts** (2 No.) ; area = R/(0.45 f_{ck}) (or masonry bearing 0.4–1 MPa); thickness for cantilever projection:

$$t = \sqrt{(2.5\ w\ (a^2 - 0.3b^2)\gamma_{m0}/f_y)} \text{ or } t = \sqrt{(3\ w\ a^2\gamma_{m0}/f_y)}$$

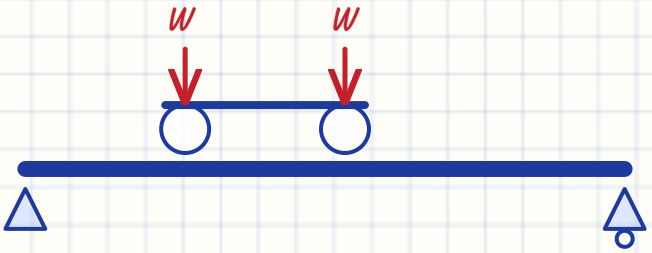
- **Roller / sliding end** (other support) for **L > 12–15 m** to allow thermal expansion — slotted holes in base plate or rollers/PTFE pads.
- Bearing on **bed block** (concrete) to spread reaction. Check **bearing stress**, anchor bolts for uplift (wind), and the bolt shear.

6. Gantry (Crane) Girder

Girder carrying an **overhead travelling (EOT) crane** wheel loads. Supported on **brackets of columns** ; span = column spacing (6–10 m typical).



Plan: rail + I-section + top channel
(channel resists lateral surge)



M_max under a wheel when span-centre bisects wheel & resultant

$M = W(2L - a)^2 / 8L$

Gantry girder section and moving wheel loads

Loads (IS 875 Part 2 / IS 800)

- **Vertical** : dead (rail ~ 0.3–0.5 kN/m, girder self-wt), **maximum wheel loads** from crane (crane + crab + lifted load) with **impact** : **25 % (EOT)**, 10 % (manual/pendant)
- **Lateral (surge)** : **10 %** of (weight of crab + lifted load) – shared by wheels on one side (EOT)
- **Longitudinal (braking)** : **5 %** of max. wheel load

Load combos: 1.5 [DL + (vertical + impact) + lateral] ; check each separately.

Section

- **Symmetrical I** (ISMB/ISWB) with **channel on top flange** (resists lateral), or plate girder for heavy cranes, or box.
- Rail fixed on top flange with clips; rail + flange act together.

Design steps

- 1. **Maximum B.M.** from moving loads (position by influence line) : for two equal wheels W at spacing a on span L:

$M_{max} = W(2L - a)^2 / (8L)$ (under one wheel when centre bisects wheel & resultant, $a < 0.586\,L$)

Max shear = reaction when wheel next to support : $V = W + W(L - a)/L$.

- 2. M_u (from vertical + impact + self-wt) and M_y (lateral, resisted by top flange + channel).
- 3. **Select section** : Z_{req}; **LTB check** with L_{LT} = span (compression flange laterally supported only at the ends, unless surge girder or bracing).
- 4. **Check combined bending**: $M_z/M_{dz} + M_y/M_{dy} \leq 1.0$.
- 5. **Shear** $V \leq V_d$ (low shear) ; **web buckling / crippling** under wheel (b_1 = wheel contact + rail dispersion at 45° through rail & flange).
- 6. **Deflection (vertical, w/o impact)** : L/750 (EOT ≤ 50 t), L/500 manual, L/1000 > 50 t ; lateral deflection L/ 1000-ish.
- 7. **Fatigue** (cyclic loads) – welded connections, bracket & **end connections to column** (bolted with bearing stiffeners).
- 8. Check **bracket** and **column** for eccentric load.

EXAMPLE 3 – MAX. B.M. FOR CRANE WHEELS

EOT crane : two wheels 100 kN each (including 25 % impact) at spacing a = 3 m on a span L = 6 m gantry girder. Check a < 0.586 L = 3.5 m ✓ (two-load rule applies).

$$M_{max} = 100 \times (2 \times 6 - 3)^2 / (8 \times 6) = 100 \times 81/48 = 168.75 \text{ kN·m}$$

Self-weight + rail 1.5 kN/m (service) → $M = 1.5 \times 6^2/8 = 6.75$ kN·m (≈ at centre).

Taking the 100 kN wheel loads as service loads (with impact) : $M_u \approx 1.5\ (168.75 + 6.75) =$ **263 kN·m**.

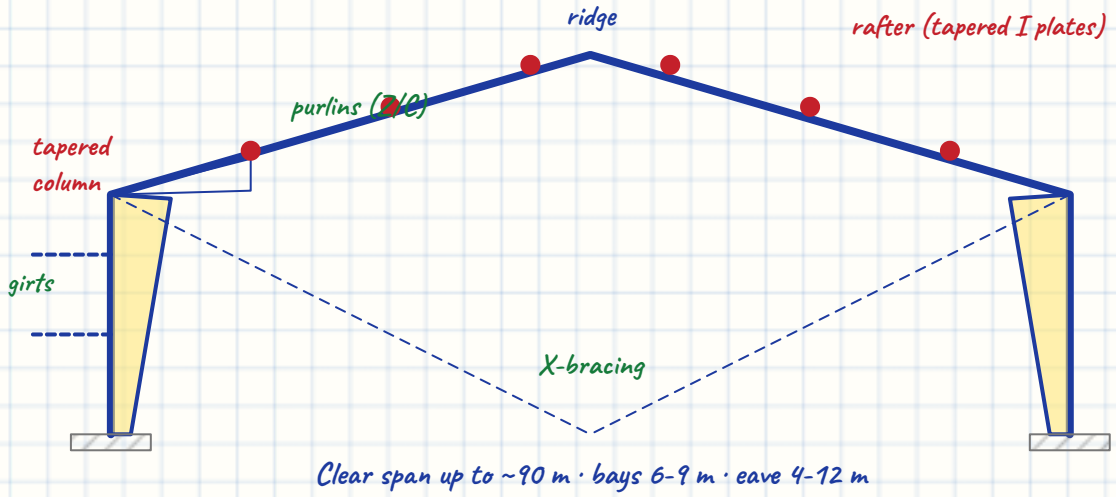
Lateral surge H = 10 % × (weight of crab + load = say 100 kN) = 10 kN per girder (shared by 2 wheels) → lateral moment $M_y \approx$ 10–15 kN·m, resisted by the top flange + channel.

Required $Z_p \geq 263 \times 10^6 \times 1.1/250 = 1157\ cm^3$ plus lateral → **ISWB 450 + ISMC 300 channel on top** (to be checked for biaxial bending & LTB).

> Maximum shear occurs with one wheel at the support; maximum deflection roughly at mid-span (wheel loads near centre).

7. Pre-Engineered Buildings (PEB)

PEB = steel buildings **designed, fabricated in the factory and bolted at site** from standard, optimised components – used for warehouses, factories, showrooms, hangars, metro stations.



PEB rigid frame with tapered members

Components

- **Primary framing** : rigid frames of **built-up tapered I-sections** (welded plates, web as thin as 5–6 mm, depth varies with bending moment) – columns, rafters, **haunch** at eave
- **Secondary framing** : **cold-formed Z / C purlins** (IS 801 / AISI), eave struts, **girts**, flange braces, sag rods
- **Roof & wall sheeting** : galvalume / colour-coated profiled sheets with insulation ; skylights, ventilators
- **Bracings** (X-bracing / rod / angle) for lateral loads ; **mezzanine floors, crane girders** can be added.
- **Foundations** : isolated footing with anchor bolts (pin or fixed base).

Advantages

- **Less steel** (≈ 30 % lighter than conventional) as depth follows moment diagram ; large clear spans (up to ~ 90 m) ; **quick erection** (weeks), pre-designed software (MBS, STAAD) ; economical, expandable ; good quality control.
- **Limitations**: limited to standard shapes; difficult for heavy crane loads (> 20–30 t) without customisation; fire protection, thin members → corrosion maintenance.

Design (outline)

- Loads : DL, LL (0.57 kN/m² typ. — IS 875) , **wind** (IS 875-3), seismic (IS 1893), crane, collateral. Combinations as IS 800 / MBMA.
- Analysis by **2-D frame software** ; design members by **IS 800 / AISC** for **web slenderness** and **tension-field action**; check deflection (**H/150 ; L/180–L/240**), local buckling.
- Connections : **end-plate moment connections with HSFG bolts** at haunch & ridge ; bolted base plate with anchor bolts.
- Codes: IS 800, IS 875, IS 801 (cold-formed), MBMA (Metal Building Manufacturers Association) manual.

Conventional vs PEB

Item	Conventional	PEB
Members	hot-rolled sections	tapered welded plates + CFS purlins
Weight	higher	25–30 % lower
Design/fabrication time	long	short, standardised
Site work	high	bolting only
Flexibility	high	moderate

8. Worked Examples – Truss

Example 4 – Wind load on a roof (IS 875 Part 3 style)

Basic wind speed V_b = 44 m/s ; $k_1 = k_2 = k_3 = k_4 = 1.0$ (assumed) → V_z = **44 m/s**

$p_z = 0.6 \times 44^2 = 1162$ N/m² = **1.16 kN/m²** ; $p_d = K_d\, p_z = 0.9 \times 1.16$ = **1.045 kN/m²** (K_a K_c taken as 1.0, conservative)

Roof slope 20° (low-rise): leeward / windward suction $C_{pe} \approx -0.7$ (read exact value from IS 875 Table) ; medium openings $C_{pi} = \pm 0.5$.

Worst net : $C_{pe} - C_{pi} = -0.7 - (+0.5) =$ **−1.2** → $F = -1.2 \times 1.045 =$ **−1.25 kN/m²** (suction, ⊥ roof).

Truss spacing 4 m, purlin spacing 1.5 m → joint (purlin) uplift = $1.25 \times 4 \times 1.5$ = **7.5 kN** per purlin ; factored $1.5 \times 7.5 = 11.3$ kN. Check 0.9 DL + 1.5 WL : **net uplift** → members in bottom chord & webs may reverse (tension ↔ compression) ⇒ design for both.

Example 5 – Member forces + design (symmetrical roof truss)

Span 12 m, rise 3 m → $\tan\theta = 0.5$, $\theta = 26.57°$, rafter length = $\sqrt{(6^2 + 3^2)} = 6.71$ m, 2 panels per rafter (3.35 m). Factored load = 1.5 kN/m² (plan) × 4 m spacing × 12 m = 72 kN → joint loads (plan panels 3 m): eave 9, mid 18, ridge 18, mid 18, eave 9 kN. Reactions R = 36 kN each.

Joint at left support (only rafter + bottom chord meet) : net vertical = 36 − 9 = 27 kN

> Rafter (end panel) : $C = 27 / \sin\theta = 27/0.447 =$ **60.4 kN (compression)**
> Bottom chord (end panel) : $T = C\cos\theta = 60.4 \times 0.894 =$ **54.0 kN (tension)**

Check by sections at mid-span : M about ridge = 36×6 − 9×6 − 18×3 = 108 kN·m → central bottom chord = 108/3 = 36 kN (T) ✓ (smaller than the end panel – consistent with a Fink/Howe web pattern).

Rafter design : P_u = 60.4 kN, L = 3.35 m, 2 ISA 65×65×6 back-to-back (A = 1488 mm², $r_{xx} \approx 19.7$ mm from tables)

$\lambda = 0.85 \times 3354 / 19.7 =$ **145** < 180 ✓ ; class c ($\alpha = 0.49$) : $f_{cc} = \pi^2E/145^2 = 93.9$; $\bar{\lambda} = 1.632$; $\varphi = 2.18$; $\chi = 0.275$

$f_{cd} = 0.275 \times 250/1.1 = 62.6$ MPa → $P_d = 62.6 \times 1488 =$ **93 kN > 60.4** ✓

Bottom chord T_u = 54 kN : $A_g(req)$ = 54×10³×1.1/250 = 238 mm² ; provide minimum **2 ISA 50×50×6** → check T_{dn}, T_{db} and slenderness (350) ✓ (governed by minimum size).

EXAMPLE 6 – BEARING PLATE FOR THE FIXED END

R_u = 36 kN on M20 concrete pedestal : $f_{bearing}$ = 0.45 × 20 = 9 MPa → A_{req} = 36 000/9 = 4000 mm² → provide practical **200 × 200** plate (area 40 000, w = 0.9 N/mm²) ; projection a = 75 mm

$t = \sqrt{(3\ w\ a^2\ \gamma_{m0}/f_y)} = \sqrt{(3 \times 0.9 \times 75^2 \times 1.1/250)}$ = **8.2 mm → 10 mm** plate ; 2 anchor bolts M16 ; roller end : slotted holes 20 × 40.

9. Bracing, Girts and Stability of the Shed

Bracings in an industrial shed

- **Roof (horizontal) bracing** – X-bracing in the plane of rafters/bottom chord at end bays & every 4th–5th bay : transfers wind from gable columns to side walls and gives lateral support to top chord (compression) → reduces L_y.
- **Vertical bracing between trusses** (at ridge, eaves) – prevents overturning/lateral buckling of trusses during erection.
- **Longitudinal bracing in columns** – resists wind/crane braking along the length.
- **Knee braces / portal action** for lateral stiffness of frame.
- Bracing members : angles, flats, rods (tension only – designed for tension, 2 rods per panel).

Girts (sheeting rails)

Horizontal members on wall columns carrying cladding & wind ⊥ to wall: designed like purlins (bending about both axes; sag rods to resist vertical self-weight/sheeting). Deflection limit L/150 (elastic) or L/180.

Column of industrial building (just idea)

- **Stepped column** with crane bracket : lower part carries crane + roof; upper part carries roof only.
- Design for axial + bending (IS 800 Cl. 9.3) ; base connected with anchor bolts (fixed/pinned).
- Wind on walls → eave reactions ⇒ horizontal load on column; frames analysed for portal action.

Angle purlin – quick checks (small spans)

- Use **ISA 100×75×8 / 90×90×8** for spans ≤ 4 m, long leg ⊥ to roof ; connect to rafter by **cleat/ angle** (bolted).
- Moment M_x about the geometric axis parallel to roof, M_y about the other : angle is **un-symmetric** → bending about principal axes (u-u, v-v); principal moment components : $M_u = M_x \cos\alpha + M_y \sin\alpha$ ...
- Use gross elastic section modulus with 1.2 limit; check deflection (≈ governing).

Channel purlin – quick checks

- ISMC 100–150 with flanges toe-up/ toe-down; shear centre issue → torsion small if sheeting connected to top flange.
- Good for spans 4–6 m; **web crippling** at the bearing on the rafter, **local bearing** with cleat.
- Cold-formed **Z / C purlins** (IS 801) with lap splices are used for spans 6–9 m (PEB).

Important formulae – Unit IV (quick sheet)

$$w_n = w\cos\theta\ ;\ w_p = w\sin\theta\ ;\ M_x = w_n L^2/10 \text{ (cont.)} / w_n L^2/8 \text{ (SS)}$$

$$p_z = 0.6\, V_z^2\ ;\ F = (C_{pe} - C_{pi})\, A\, p_d$$

$$\text{LL on roof} = 0.75 - 0.02(\theta - 10)\ \text{kN/m}^2 \geq 0.4$$

$$\text{Joint load} = (\text{kN/m}^2) \times \text{truss spacing} \times \text{purlin spacing (plan)}$$

$$M_{max} = W(2L - a)^2/(8L) \text{ (2 equal wheel loads)}\ ;\ \text{vertical impact 25 \% (EOT)}$$

$$\text{Surge} = 10\ \%\ (\text{crab} + \text{load})\ ;\ \text{Longitudinal braking} = 5\ \%\ \text{wheel load}$$

$$t_{base} = \sqrt{(3\, w\, a^2 \gamma_{m0} / f_y)}\ \text{(cantilever base plate)}$$

Important exam questions

1. Explain different types of roof trusses, with neat sketches.
2. Calculate wind load on a pitched roof truss (IS 875) and panel-point loads.
3. Design an angle purlin for a roof with given loads / design a channel purlin with sag rods.
4. Design a Fink truss member (rafter & tie) and the welded joint at the heel.
5. Explain loads on gantry girders and design a gantry girder for a 10 t EOT crane.
6. Write short note on PEB – components, advantages, design considerations.

UNIT – V

Plastic Analysis and Design

Syllabus: Introduction to Plastic Analysis – Theory of Plastic Analysis – Design of Continuous Beams and Portal Frames Using Plastic Design Approach.

What's inside these notes

- Elastic vs plastic behaviour, ductility, assumptions
- Plastic moment M_p, plastic section modulus, shape factor
- Plastic hinge, collapse mechanisms, load factor
- Theorems: static, kinematic (upper/lower bound), uniqueness
- Methods: virtual work (mechanism) & statical method
- Plastic design of continuous beams (examples)
- Plastic design of portal frames – beam, sway & combined mechanisms

Exam tip: Plastic collapse problems: (1) find hinge positions, (2) write virtual work equation for EACH mechanism, (3) the LEAST collapse load (or the GREATEST M_p required) governs, (4) check M ≤ M_p everywhere.

Codes used: IS 800:2007 (Limit State Method), IS 808 / SP-6(1) steel tables, IS 875 (loads), IS 2062 (steel grades). Take E = 2×10^5 MPa, f_y = 250 MPa, f_u = 410 MPa unless stated.

1. Introduction to Plastic Analysis

Elastic design limits stress at the extreme fibre to f_y (first yield). But ductile steel can carry more load: after the extreme fibres yield, the **yielded zone spreads inwards**, finally the whole section is plastic and the structure **collapses** only after enough **plastic hinges** form to make a **mechanism**.

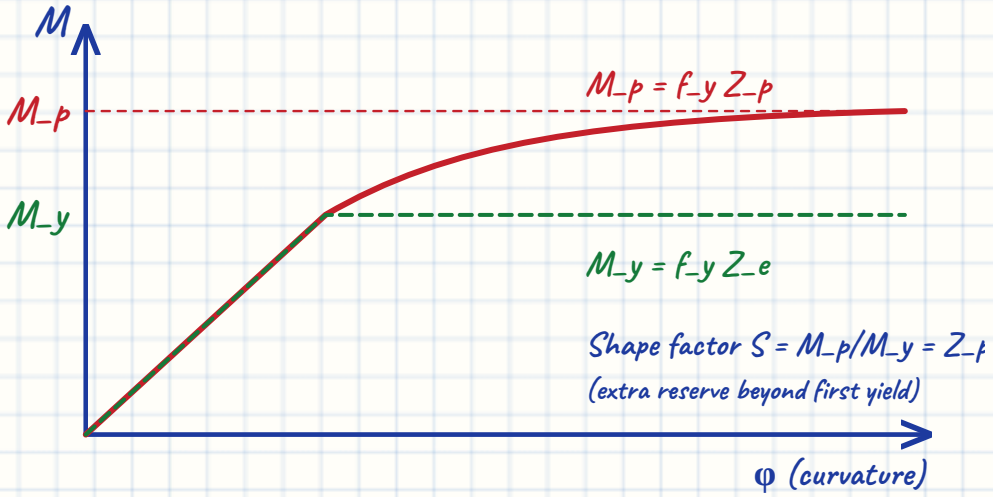
Plastic analysis finds the **collapse load (ultimate load)** directly, uses the full strength reserve and gives a more rational **load factor** $\lambda = W_c$ / W_working.

Why plastic analysis?

- More **economical** (10–30 % saving, esp. in continuous beams/frames)
- Gives the **true load factor** against collapse; redistribution of moments is accounted for
- Simple (statics only, no compatibility/ stiffness needed)
- Applicable to **ductile** steel (E250 ... E450) with **plastic or compact sections**.

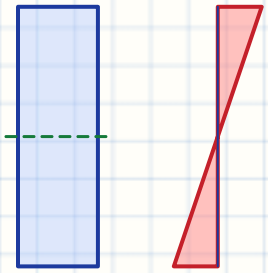
Assumptions

- Stress–strain curve is **idealised** elastic–perfectly plastic ($\sigma = f_y$ after yield) ; strain hardening ignored.
- **Plane sections remain plane** after bending ; tension = compression behaviour.
- Material is **homogeneous & isotropic** ; ductile (large strain without fracture)
- **Axial & shear effects** on M_p neglected (or corrected) ; local & lateral buckling prevented (adequate bracing, compact sections) ; stable under deflection (**small deformation** theory) ; loads are **proportional & static** (increase together).
- Connections are **full strength** & able to transmit M_p.



Moment–curvature relation (idealised)

2. Plastic Moment & Shape Factor



Stress distribution as the moment increases

- **Elastic** : $f = M\,y / I$; extreme fibre = f_y at **yield moment** $M_y = f_y\,Z_e$
- **Elasto-plastic** : yielded zones spread in from the extreme fibres (elastic core remains)
- **Fully plastic** : all fibres at f_y (tension below, compression above) ⇒ **plastic moment** (limit of capacity)

$$M_p = f_y\,Z_p\,,\; M_y = f_y\,Z_e$$

Plastic neutral axis

At full plasticity the **N.A. divides the area into two EQUAL halves** (so that C = T) — the **equal-area axis** (differs from elastic centroidal axis for un-symmetric sections).

$$Z_p = (A/2)(\bar{y}_1 + \bar{y}_2) = A_c\,y_c + A_t\,y_t$$ (y = distance of centroid of each half-area from the equal-area axis)

Shape factor

$$S = M_p / M_y = Z_p / Z_e$$

Section	S
Rectangle	**1.50**
Solid circle	1.70
Diamond (square on diagonal)	2.00
Triangle	2.34
I-section / channel (rolled)	**1.10 – 1.20** (≈1.14)
Thin circular tube	1.27
T-section	1.8 (approx.)

Higher S → more reserve beyond yield. For I-sections the flanges give most Z_e, so S is small.

EXAMPLE 1 – RECTANGLE B × D

$Z_e = bd^2/6$; $Z_p = 2 \times (bd/2) \times (d/4) = bd^2/4$ ⇒ **$S = (bd^2/4)/(bd^2/6) = 1.5$** ; $M_p = f_y\,b\,d^2/4$.

Example 2 – Built-up I-section : flanges 150×20, web 10×300 (total depth 340)

$I_z = [150\times340^3 - 140\times300^3]/12 = [5.8956\times10^9 - 3.78\times10^9]/12 = 1.763\times10^8\ mm^4$; $Z_e = I/170 = \mathbf{1.037\times10^6\ mm^3}$

Equal-area axis = centroidal axis (symmetric) → $Z_p = 2[150\times20\times160] + 2[10\times150\times75] = 960\,000 + 225\,000 = \mathbf{1.185\times10^6\ mm^3}$

$$S = 1.185/1.037 = 1.14$$

$M_p = 250 \times 1.185\times10^6 = \mathbf{296\ kN\cdot m}$; $M_y = 259\ kN\cdot m$

Example 3 – T-section (equal-area axis)

T : flange 100 × 10 (top), web 10 × 90 (total depth 100) ; $A = 1000 + 900 = 1900\ mm^2$ → half = $950\ mm^2$.

Flange area 1000 > 950 ⇒ **plastic N.A. lies inside the flange**, at $y = 950/100 = \mathbf{9.5\ mm}$ **from the top** (the elastic N.A. is at 28.7 mm – they differ!).

$Z_p = \Sigma\, A\, y$ (each part about the plastic axis) $= 950 \times 4.75 + 50 \times 0.25 + 900 \times 45.5 = 4512 + 12 + 40\,950 = \mathbf{45\,475\ mm^3}$

Elastic : $\bar{y} = (1000\times5 + 900\times55)/1900 = 28.7\ mm$; $I = 1.80\times10^6\ mm^4$; $Z_e = I/71.3 = 25\,240\ mm^3$

$$S = 45\,475/25\,240 = 1.80$$

Example 4 – Statical method for the propped cantilever (UDL w)

Moment at fixed end A : $M_A = -M_p$ ⇒ taking moments about A : $R_B\, L - wL^2/2 = -M_p$ ⇒ $\mathbf{R_B = wL/2 - M_p/L}$.

Sagging moment at distance x from the prop B : $M = R_B\, x - w\, x^2/2$; maximum where shear = 0 ⇒ $\mathbf{x = R_B/w}$, $M_{max} = R_B^2/(2w)$.

Plastic condition $M_{max} = M_p$ ⇒ $(wL/2 - M_p/L)^2 = 2\, w\, M_p$. Let $k = wL^2/M_p$: $(k/2 - 1)^2 = 2k$ ⇒ $k^2 - 12k + 4 = 0$ ⇒ $\mathbf{k = 6 + \sqrt{32} = 11.66}$

$$w_c = 11.66\, M_p/L^2 \checkmark \text{ (same as the kinematic result) and } x = R_B/w = 0.414\, L$$
$$\text{from the prop} \Rightarrow \text{upper bound = lower bound = true collapse load.}$$

Example 5 – Shape factor of a solid circle (diameter d)

$Z_e = \pi\, d^3/32$; plastic N.A. is the diameter, each half-area $= \pi\, d^2/8$ with centroid at $2d/(3\pi)$ from the axis

$Z_p = 2 \times (\pi d^2/8) \times (2d/3\pi) = \mathbf{d^3/6}$ ⇒ $S = (d^3/6)/(\pi d^3/32) = 16/(3\pi) = \mathbf{1.70}$.

3. Plastic Hinge & Mechanisms

Plastic hinge = a section where $M = M_p$ – the section rotates (plastic rotation) at **constant moment M_p** like a hinge with a friction torque. Differences from a real hinge:

carries M_p ; **unidirectional** (rotation in direction of M only) ; has a small finite length in reality but is idealised at a point.

- **Collapse mechanism** : enough hinges form so that the structure becomes a **kinematic chain (mechanism)** – deflections grow without further load increase.
- Number of hinges required for complete collapse : **n = r + 1** (r = **degree of static indeterminacy**) for a **complete** collapse ; fewer hinges (**partial / over-complete collapse**) in some cases.



● = plastic hinge (r+1 hinges for collapse; r = degree of indeterminacy)

Collapse mechanisms of beams

Locations of plastic hinges

- Points of **maximum moment** : under concentrated loads, fixed/continuous supports, at joints of frames (corners), within spans for UDL (where shear = 0).
- Hinge at a **joint of 3 or more members** forms in the weakest member.

Load factor

λ (load factor) = Collapse load W_c / Working load W

IS 800:2007 works with **factored loads** (Table 4, e.g. 1.5 (DL + LL)) and design $M_d = Z_p f_y/\gamma_{m0}$ (older IS 800:1984 used a load factor of **1.7** for DL + LL, 1.3–1.4 with wind).

Mechanisms in frames

- **Beam mechanism** – hinges in beam (3 hinges)
- **Panel / sway mechanism** – columns sway (4 hinges: tops & bottoms of columns)
- **Gable mechanism** – rafter / ridge of gable frames
- **Joint mechanism** – rotation at joint of 3+ members (rarely)
- **Combined mechanism** – combination of independent mechanisms to eliminate a hinge → **lowest collapse load**.

Number of independent mechanisms n = N – r (N = number of possible hinge locations ; r = degree of redundancy).

4. Theorems of Plastic Collapse

For a proportionally increasing load with a given structure & M_p :

- **Static (Lower bound) theorem** – A load W_s for which a **statically admissible moment diagram** exists (equilibrium + $M \leq M_p$ everywhere) is **≤ actual collapse load** (**safe** estimate).

- **Kinematic (Upper bound) theorem** – A load W_k computed for **any assumed mechanism** (by virtual work) is **≥ actual collapse load** (**unsafe** estimate).
- **Uniqueness theorem** – The collapse load W_c is the one for which BOTH are satisfied : the moment diagram is statically admissible **and** the mechanism is formed ($W_s = W_k$).

$$W_s \le W_c \le W_k$$

So **try all possible mechanisms** → collapse load = **smallest** W_k; then verify the **B.M. diagram** (statical check) to make sure no $|M| > M_p$.

Methods of analysis

- **Kinematic (mechanism) method** – virtual work: **External work = Internal work**

$$\Sigma W\delta = \Sigma M_p \theta$$

δ = virtual displacement at load, θ = virtual hinge rotation. Internal work is always positive : $M_p \times |\theta|$. At a hinge between two members rotating by θ_1, θ_2 : relative rotation.

- **Statical method** – draw free + reactant B.M.D. : assume redundant, put $M = \pm M_p$ at the hinges and use equilibrium to find W.
- **Moment-balancing / Instantaneous centre method** (frames).

5. Collapse Load of Beams (Standard Results)

Let M_p = plastic moment, span L. **Virtual work** : $W \times \delta = M_p \Sigma \theta$.

(a) Simply supported, central point load W

Hinge at centre; $\delta = (L/2)\ \theta$; rotation at hinge = 2θ : $W\ (L/2)\ \theta = M_p\ (2\theta)$ →

$$W_c = 4\ M_p / L \text{ (UDL : } w_c = 8\ M_p / L^2)$$

(b) Fixed beam, central point load

Hinges at both ends and centre : rotations θ, 2θ, θ (total 4θ) : $W\ (L/2)\ \theta = 4\ M_p\ \theta \Rightarrow$ **$W_c = 8\ M_p / L$**

UDL : work = $w \times (½ \cdot L \cdot L\theta/2) = w\ L^2\ \theta/4 = 4\ M_p\ \theta \Rightarrow$ **$w_c = 16\ M_p / L^2$**

(c) Propped cantilever (fixed A, prop at B), central point load

Hinges at A and under the load : rotations θ and 2θ (total 3θ) → $W\ (L/2)\ \theta = 3\ M_p\ \theta \Rightarrow$ **$W_c = 6\ M_p/L$**

UDL : hinge at A and a sagging hinge at **0.586 L from A (= 0.414 L from the prop)** →

$$w_c = 11.66\ M_p / L^2\ (= 2(3 + 2\sqrt{2})\ M_p/L^2)$$

(d) Cantilever, end load : $W_c = M_p / L$.

Beam	Hinges	W_c (point at centre)	w_c (UDL)
Simply supported	1	$4M_p/L$	$8M_p/L^2$
Fixed–fixed	3	$8M_p/L$	$16M_p/L^2$
Propped cantilever	2	$6M_p/L$	$11.66M_p/L^2$
Cantilever (end load)	1	M_p/L	—

Example 6 – Propped cantilever with UDL (derivation)

Fixed end A, prop B. Hinges at A and at C, a distance x from A. Let AC rotate θ about A, CB rotate φ about B ; deflection at C : $\delta = x\theta = (L - x)\varphi \Rightarrow \varphi = x\theta/(L - x)$.

Hinge rotations : $A = \theta$; $C = \theta + \varphi = \theta\, L/(L - x)$; $B = 0$ (pin).

Internal work $= M_p\,[\theta + \theta\, L/(L - x)] = M_p\,\theta\,(2L - x)/(L - x)$

External work $= w \times \tfrac{1}{2}\, L\, \delta = \tfrac{1}{2}\, w\, L \times \theta$

$$w = 2\,M_p\,(2L - x)\,/\,[L \times (L - x)]$$

Minimise w.r.t. x : $x^2 - 4Lx + 2L^2 = 0 \Rightarrow$ **$x = (2 - \sqrt{2})L = 0.586\,L \Rightarrow w_c = 11.66\,M_p/L^2$**

6. Plastic Design of Continuous Beams

Idea: For the **factored loads**, find the collapse mechanism of each span and the required M_p ; select a section with $Z_p \geq M_p\,\gamma_{m0}/f_y$ (IS 800: $M_d = Z_p\, f_y/\gamma_{m0} \geq M_u$).

Possible mechanisms in a continuous beam

- Each span can fail **independently** (a span between two hogging hinges + one sagging hinge = **fixed-beam mechanism**) ; end spans (propped-cantilever type) ; **combined** mechanisms are not possible across supports (no relative displacement at supports).
- For a **uniform M_p**, consider every span separately & take the **largest required M_p**.

Steps

- 1. Factored loads on each span (1.5 (DL + LL) or the relevant IS 800 Table 4 combination).
- 2. Choose mechanism for each span : **end span** with simple support → **propped-cantilever** (hinge near 0.414 L from the outer support : $w\,M_p = 11.66\ldots$), **interior span** both ends continuous → **fixed-beam type** ($w = 16\,M_p/L^2$).
- 3. Compute $M_p(\text{req}) = W\,L\,/\,k$ for each span. Take **max** ; if different sections per span economical use span-wise M_p (but hinges at supports use the **smaller** M_p of the two adjacent spans).
- 4. **Select $Z_p = M_p\,/\,(f_y/\gamma_{m0})$** ; check shape factor/class (plastic), shear $V \leq 0.6\,V_d$, web buckling, **lateral bracing**, deflection at working load.
- 5. **Check the B.M. diagram** : $|M| \leq M_p$ everywhere.

Example 7 – Two-span continuous beam, equal spans L = 6 m, UDL w_u = 40 kN/m

FACTORED

Each span is like a propped cantilever (hinge over middle support + sag hinge at 0.414 L from the outer support... both spans identical).

$$w_u = 11.66\, M_p/L^2 \Rightarrow M_p = 40 \times 6^2/11.66 = 123.5\ kN\cdot m$$

$Z_p(req) = 123.5\times10^6 \times 1.1/250 = 543\,600\ mm^3 =$ **544 cm³** → ISMB 300 ($Z_p \approx 654\ cm^3$) ✓ ; ISMB 250 ($Z_p \approx 467$) ×.

Check shear : $R_A = wL/2 - M_p/L = 120 - 20.6 = 99.4$ kN ; total at inner support = $2(240 - 99.4) = 281$ kN ⇒ V_u (each side) = **140.6 kN** < $0.6\ V_d$ (≈ 182 kN for ISMB 300) ✓

Provide ISMB 300 (with lateral bracing for compression flange near supports).

EXAMPLE 8 – CONTINUOUS BEAM, SPANS 5 M AND 6 M WITH DIFFERENT LOADING (OUTLINE)

Span AB = 5 m, central point load W_1 = 100 kN (factored) ; BC = 6 m, UDL w = 30 kN/m (factored), A, C simply supported.

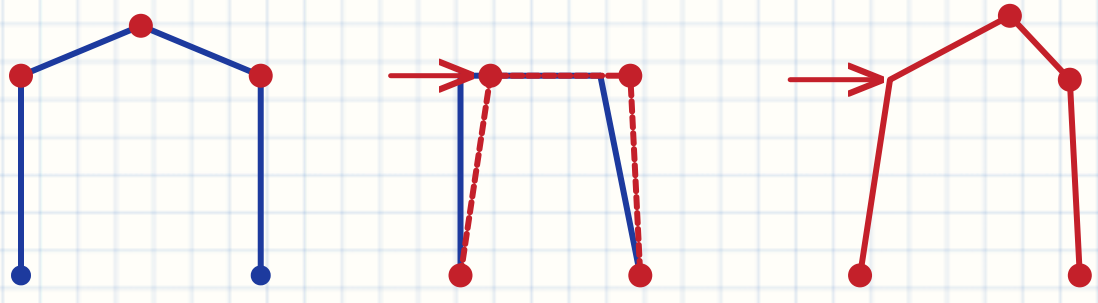
Mechanism AB (propped cantilever with point load at centre) : $W_1 = 6\ M_p/L \Rightarrow M_p = 100\times5/6 =$ **83.3 kN·m**

Mechanism BC (propped cantilever with UDL) : $w = 11.66\ M_p/L^2 \Rightarrow M_p = 30 \times 36/11.66 =$ **92.6 kN·m** (governs)

Hence **M_p = 92.6 kN·m → Z_p = 407 cm³ ⇒ ISMB 250** ($Z_p \approx 467\ cm^3 \geq 407$) and check the B.M.D. (no moment above M_p).

7. Plastic Analysis of Portal Frames

Fixed-base single-bay portal frame (columns h, beam L, **uniform M_p**, vertical load W at mid-span of beam, horizontal H at eaves level). Possible plastic hinge positions : **A, E (bases), B, D (eaves), C (under W)** → **N = 5**, r = 3 ⇒ **independent mechanisms n = 5 – 3 = 2** (beam, sway) and **1 combined**.



Beam, sway & combined mechanisms of a portal frame

Virtual work equations (θ = virtual rotation)

- **Beam mechanism** : hinges at B, C, D (θ, 2θ, θ) ⇒ $4M_p\theta = W\ (L/2)\ \theta \Rightarrow W = 8\ M_p / L$

- **Sway mechanism** : hinges at A, B, D, E (θ each), eaves displace hθ ⇒ 4M_pθ = H h θ ⇒ **H = 4 M_p / h**
- **Combined mechanism** : eliminate hinge at B (rotations θ and −θ cancel) : hinges A(θ), C(2θ), D(2θ), E(θ) → 6M_pθ

$$6\,M_p = H\,h + W\,L/2$$

(Hinge at D rotates θ + θ = 2θ, because the sway rotation and the beam rotation add.)

> With only vertical loads the combined and sway mechanisms vanish → **beam mechanism**. With a fixed-base frame, pinned bases give hinge at A, E removed (3 hinges in sway; M_p required increases).

Procedure : compute M_p from each mechanism → **maximum M_p (least collapse load)** governs → check the **moment diagram** (equilibrium) and **column design for axial + M_p**.

EXAMPLE 9 – FIXED-BASE PORTAL FRAME : H = 4 M, L = 6 M, W = 100 KN (MID-SPAN), H = 50 KN (EAVES) – FACTORED

Beam : W (L/2) θ = 4 M_p θ ⇒ M_p = W L / 8 = 100 × 6 / 8 = **75 kN·m**

Sway : H h = 4 M_p ⇒ M_p = 50 × 4 / 4 = **50 kN·m**

Combined : 6 M_p = H h + W L/2 = 50 × 4 + 100 × 3 = 500 ⇒ M_p = **83.3 kN·m** ← largest

Required **M_p = 83.3 kN·m** (combined mechanism governs – the **least collapse load**).

Z_p(req) = 83.3×10⁶ × 1.1/250 = **367 cm³** → choose ISMB 250 (Z_p ≈ 1.14 × 410 = 467 cm³) or equivalent.

Moment check (statics) : with hinges A, C, D, E at ±M_p, equilibrium of the beam and columns gives M_B (the hinge eliminated in the combined mechanism) — verify $|M_B| \le M_p$; if so, upper bound = lower bound and this is the true collapse mechanism.

Effect of axial force & shear on M_p

- Axial load N reduces plastic moment : **$M_{pc} = M_p\,[1 - (N/N_y)^2]$** (rectangle) ; for I-section, if $N \le 0.15\,N_y$ neglect, else use reduced M_{pc} (see IS 800 Cl. 9.3).
- High shear ($V > 0.6\,V_d$) reduces M_p (see Unit III).
- Columns also checked for **buckling** with plastic M_p; ensure **adequate bracing, stiff joints**, and **rotation capacity** (plastic or compact sections only).

8. Gable Frames, Instantaneous Centres & Checks

Gable frame mechanism

Rafters inclined; hinges at eaves, ridge (apex) and bases. Use **instantaneous centres (I.C.)** of each rigid link to relate rotations and displacements.

- Beam (rafter pair) mechanism: rafter rotates about I.C. ; sway : eaves displace horizontally; combined etc.
- Virtual work with horizontal & vertical displacement of load points: δ_H = (distance of load point from I.C.) × θ etc.

Moment-balance (statical) check of portal frame

After finding M_p from the critical mechanism, check:

- 1. Draw free B.M.D. (as for simply supported beam) and reactant diagram from M_A, M_E, M_B, M_D... with chosen redundants (H_A, M_A).
- 2. Superpose ; verify **$|M| \le M_p$ everywhere**. If any $M > M_p$, another mechanism governs.
- 3. Equilibrium equations ΣM, ΣH, ΣV hold.

Advantages / limitations of plastic design

Advantages	Limitations
Economical, uses reserve strength	Only for ductile steel & **compact sections**
Simple analysis (statics)	Deflection & serviceability must be checked separately (working loads)
True load factor	Local/lateral buckling, fatigue & fracture must be controlled
Redistribution captured	Repeated / alternating loads may cause incremental collapse

EXAMPLE 10 – FIXED BEAM, L = 8 M, UDL w_u = 30 KN/M (FACTORED)

$w_c = 16\, M_p/L^2 \Rightarrow$ **$M_p = 30 \times 8^2/16 = 120$ kN·m** (hogging at both ends & sagging at centre, each = M_p).

$Z_p(req) = 120\times10^6 \times 1.1/250 =$ **528 cm³** → ISMB 300 ($Z_p \approx 654$) ✓.

(Elastic design needs $M = wL^2/12 = 160$ kN·m at the supports $\Rightarrow Z_e = 160\times10^6\times1.1/250 = 704$ cm³ ⇒ ISMB 350 – plastic design saves one size.)

EXAMPLE 11 – SAME PORTAL FRAME BUT WITH PINNED BASES (H = 4, L = 6, W = 100, H = 50)

Hinge positions : B, C, D only ($N = 3$, $r = 1 \Rightarrow n = 2$ mechanisms).

Beam : $4\, M_p = W\, L/2 \Rightarrow$ **M_p = 75 kN·m**

Sway : hinges at B, D only : $2\, M_p\, \theta = H\, h\, \theta \Rightarrow M_p = 50\times4/2 =$ **100 kN·m**

Combined : eliminate B ; C (2θ), D (2θ) : $4\, M_p\, \theta = H\, h\, \theta + W\, (L/2)\, \theta \Rightarrow M_p = (200 + 300)/4 =$ **125 kN·m** ← governs

Pinned bases need M_p = 125 against 83.3 for fixed bases (≈ 50 % more steel in the frame; foundations are cheaper, though).

9. Rotation Capacity, Redistribution & Practical Points

- **Rotation capacity** $R = (\theta_u - \theta_p)/\theta_p$: ability of the hinge to rotate at M_p without losing strength. It needs **plastic (Class 1) sections** ; limits : flange ≤ 9.4 ε, web ≤ 84 ε (rolled I) – see Unit I.
- **Moment redistribution** : in an indeterminate structure, when the first hinge forms, extra load is carried by the remaining sections until the mechanism is complete – this is the extra strength plastic design uses.

- **Lateral bracing** : braces close to hinge locations so LTB does not occur before the hinge rotates.
- **Connections** must be **stronger** than the members (full-strength, haunches or stiffeners at knees).
- **Incremental collapse (ratcheting)** & **alternating plasticity** under repeated loads – shakedown theory (awareness only).
- **Deflection** is checked at **working** (unfactored) loads – elastic.

EXAMPLE 12 – THREE-SPAN CONTINUOUS BEAM, EQUAL SPANS 5 M, UDL 36 KN/M (FACTORED)

End spans (propped-cantilever type) : $M_p = w\,L^2/11.66 = 36 \times 25/11.66 =$ **77.2 kN·m**

Interior span (both ends continuous – fixed-beam type) : $M_p = w\,L^2/16 = 36 \times 25/16 =$ **56.3 kN·m**

Uniform section ⇒ take the **larger** : $M_p = 77.2$ kN·m ⇒ $Z_p = 77.2\times10^6 \times 1.1/250 =$ **340 cm³** → ISMB 250 ($Z_p \approx 467$) ✓.

(The interior span is under-stressed – its sagging hinge does not form first; hogging hinges at supports are shared.)

Plastic section modulus of a doubly symmetric I-section (formula)

$$Z_p = b\,t_f\,(d - t_f) + t_w\,(d - 2\,t_f)^2 / 4 \quad (d = \text{overall depth},\ b = \text{flange width})$$

Check ISMB 300 : $140\times13.1\times286.9 + 7.7\times273.8^2/4 = 526\,185 + 144\,310 \approx 670$ cm³ (tables give ≈ 654 cm³ because of root fillets/tapered flanges – always use SP-6 values).

Effect of axial force on M_p (columns of frames)

- For a **rectangle** : $M_{pc} = M_p\,[1 - (N/N_y)^2]$, $N_y = A\,f_y$.
- For **I-sections** (major axis) the reduction is small : $M_{pc} \approx 1.18\,M_p\,(1 - N/N_y) \le M_p$; **neglect** it when $N \le 0.15\,N_y$.
- Columns of frames are checked for **combined axial + bending** and **buckling** after M_p is fixed.

Elastic design vs plastic design

Feature	Elastic (WSD)	Plastic
Basis	first yield f_y at extreme fibre	formation of mechanism
Capacity	$M_y = f_y\,Z_e$	$M_p = f_y\,Z_p$ (≈ 14 % more for I)
Analysis	indeterminate (compatibility)	statics + virtual work
Safety measure	factor of safety on stress	load factor on collapse
Economy	less	10–30 % saving (continuous beams/frames)
Applicable to	any steel	ductile steel, compact sections

> Exam trick: in portal-frame questions always write **three equations** (beam, sway, combined) – then state **the greatest M_p** and say "combined mechanism governs, so M_p = … ".

Important formulae – Unit V

$M_p = f_y Z_p$; $M_y = f_y Z_e$; $S = Z_p/Z_e$ ($I \approx 1.14$, rectangle 1.5, circle 1.7)

SS beam : $W_c = 4M_p/L$; $w_c = 8M_p/L^2$

Fixed beam : $W_c = 8M_p/L$; $w_c = 16M_p/L^2$

Propped cantilever : $W_c = 6M_p/L$; $w_c = 11.66M_p/L^2$

Portal : beam $4M_p\theta = W(L/2)\theta$; sway $4M_p = Hh$; combined $6M_p = Hh + WL/2$

Hinges for collapse = $r + 1$; independent mechanisms = $N - r$; $\Sigma W\delta = \Sigma M_p\theta$

Important exam questions

1. Define plastic hinge, shape factor, load factor; state assumptions of plastic theory.
2. Derive M_p and S for rectangular & I-section.
3. State and prove the static, kinematic and uniqueness theorems.
4. Find the collapse load of a propped cantilever / fixed beam with UDL.
5. Design a two-span continuous beam by plastic theory.
6. Analyse a fixed-base portal frame – beam, sway, combined mechanisms ; select the section.